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RAFT Agent Design and Synthesis

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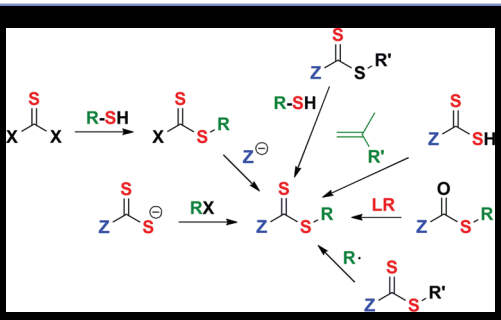
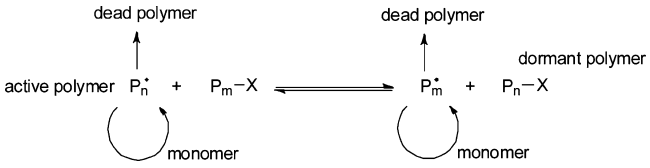
ABSTRACT:



INTRODUCTION

RAFT (reversible addition−fragmentation chain transfer) polymerization, a reversible deactivation radical polymerization (RDRP),[1] is one of the most effective and versatile methods for providing living characteristics to radical polymerization.[2−9] RAFT provides reversible deactivation of propagating radicals by degenerate chain transfer[1] for which a general mechanism is shown in Scheme 1.[4] The chain transfer step has been termed degenerate because the process involves an exhange of functionality and the only distinction between the species on the two sides of the equilibrium is molar mass.

Scheme 1. Polymerization with Reversible Deactivation by Degenerate Chain Transfer



Reports of radical addition−fragmentation processes first appeared in the synthetic organic chemistry literature in the early 1970s.[10,11] Well-known examples include allyl transfer reactions with allyl sulfides[12] and stannanes (the Keck reaction)[13] and the Barton−McCombie deoxygenation process with xanthates.[14] The use of (irreversible) addition−fragmentation chain transfer agents, such as vinyl ethers and allyl sulfides, to control molecular weight and end-group functionality of polymers was reported in the 1980s.[15−18] However, the direct use of addition−fragmentation chain transfer agents to provide living character to radical polymerization did not appear until the mid-1990s. The first RAFT agents (though that term was not used at the time) were macromonomers of general structure **1**.[19−21] The RAFT process making use of thiocarbonylthio compounds **2** to control radical polymerization appeared in 1998.[22−24] When the thiocarbonylthio compound is a xanthate (Z = *O*-alkyl), the process is also known as MADIX (macromolecular design by interchange of xanthate).[25]

1 **2**

The intention of this Perspective is to review and provide guidance on the design and synthesis of thiocarbonylthio RAFT agents (**2**). These compounds include dithioesters (Z = alkyl or aryl), trithiocarbonates (Z = SR′), xanthates (Z = OR′), and dithiocarbamates (Z = NR′R″). The question of how to choose or design the right RAFT agent for the monomer(s), the reaction conditions, and the desired functionality in the product will be addressed. We will then critically assess the various available methods for RAFT agent (and macro-RAFT agent) synthesis. Finally, we will point to some of the remaining challenges in RAFT agent design and synthesis.

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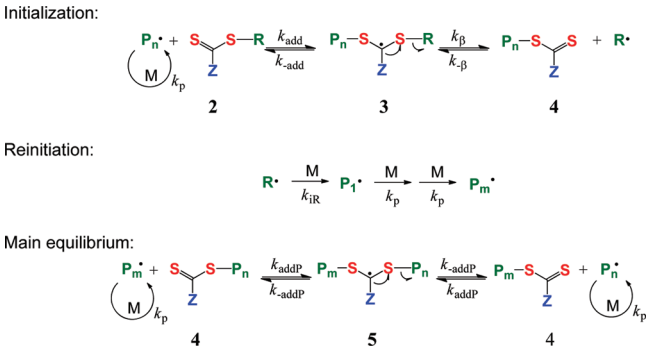
■ RAFT AGENT DESIGN AND SYNTHESIS

RAFT polymerization comprises the addition−fragmentation equilibria shown in Scheme 2 plus all of the usual processes that make up radical polymerization, most notably initiation and termination.[23] Note that radicals are neither formed nor destroyed by the RAFT steps. Thus, RAFT polymerization will not take place without an external supply of radicals from an initiator. Therefore, like ideal chain transfer reactions, the RAFT equilibria need have no direct influence on the rate of polymerization beyond that caused by the reduction in molar mass and the narrowing of the molar mass distribution. It should also be noted that radical−radical termination is not directly suppressed by the RAFT process. Although the basic mechanism shown in Scheme 2 is generally not disputed, there is ongoing debate on the detailed kinetics of the RAFT process, the rapidity with which the various equilibria are established, and what side reactions might occur to complicate the process in specific circumstances.[26−30] If fragmentation is slow, the intermediate species (**3** or **5**) is consumed in side reactions, or reinitiation is slow or inefficient, and then retardation or inhibition can result.[31]

Scheme 2. RAFT Equilibria



Optimal control in RAFT polymerization requires choosing an appropriate RAFT agent (**2**) for the monomer(s) to be polymerized and the reaction conditions. The Z and R groups both play critical roles in determining the outcome of polymerization. By determining the rate of addition and fragmentation, they control efficiency of chain transfer and the likelihood of retardation or inhibition.

The properties of RAFT agents **2** can be defined in terms of two transfer coefficients C_{tr} ($=k_{tr}/k_p$) and C_{-tr} ($=k_{-tr}/k_{iR}$) where the rate constants k_{tr} and k_{-tr} are defined in terms of the rate constants for radical addition, k_{add} and $k_{-\beta}$, and a partition coefficient (ϕ) as shown in eqs 1−3,[32−34] where the various rate constants are defined in Scheme 2.

$$k_{tr} = k_{add}\phi = k_{add}\frac{k_\beta}{k_{-add} + k\beta} \tag{1}$$

$$k_{-tr} = k_{-\beta}(1 - \phi) = k_{-\beta}\frac{k_{-add}}{k_{-add} + k_\beta} \tag{2}$$

$$\phi = \frac{k_\beta}{k_{-add} + k_\beta} \tag{3}$$

The partition coefficient ϕ indicates the preference for the intermediate radicals **3** (or **5**) to fragment to products or return to starting materials. For effective RAFT agents **2**, R should be a good homolytic leaving group with respect to the propagating radical (i.e., ϕ should be >0.5). For macro-RAFT agents **4** formed in RAFT homopolymerization, where n and $m > 2$, then $C_{tr} = C_{-tr}$ and ϕ will be ∼0.5.

The predicted dependence of the degree of polymerization and dispersity of the polymer formed on monomer conversion and the transfer coefficient ($C_{tr} = C_{-tr}$) for an ideal polymerization (no termination) with reversible chain transfer is shown in Figure 1.[33] The predicted degree of polymerization is simply the ratio of [monomer consumed]:[RAFT agent consumed]. A higher molecular weight than would be predicted with complete utilization of the transfer agent need not reflect some form of "hybrid behavior" as is suggested in some papers. It can simply indicate a low C_{tr} and that the initial RAFT agent is not fully converted to macro-RAFT agent.[31] The characteristics often associated with living polymerization, namely, the straight line dependence of molar mass on conversion (and a low dispersity, $Đ < 1.2$), require a C_{tr} of at least 10. The more effective RAFT agents have $C_{tr} > 100$.

The rate of consumption of the initial transfer agent (**2**) is given by eq 4:

$$\frac{d[\mathbf{2}]}{d[M]} \approx C_{tr}\frac{[\mathbf{2}]}{[M] + C_{tr}[\mathbf{2}] + C_{-tr}[\mathbf{4}]} = C_{tr}\frac{[\mathbf{2}]}{[M] + C_{tr}[\mathbf{2}] + C_{-tr}([\mathbf{2}]_0 - [\mathbf{2}])} \tag{4}$$

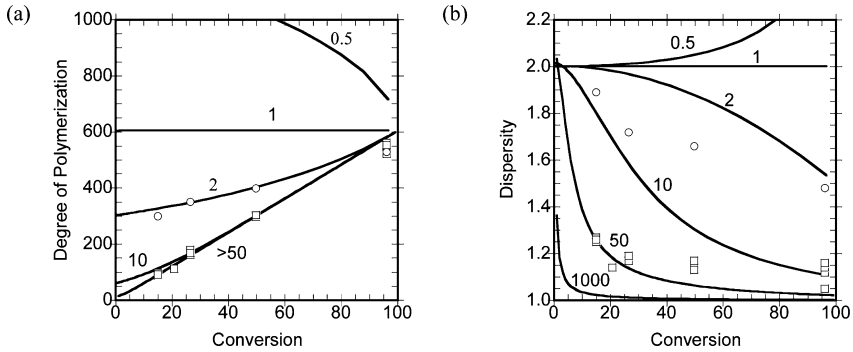


Figure 1. Predicted dependence of (a) the degree of polymerization and (b) the dispersity on conversion in polymerizations involving reversible chain transfer as a function of the chain transfer coefficient (C_{tr}). Predictions are based on equations proposed by Müller et al.[35,36] with the concentration of active species = 10^{-7} M, C_{tr} as indicated and the ratio of monomer to transfer agent = 605. Experimental data points shown are for methyl methacrylate (7.02 M) polymerization in presence of dithiobenzoate esters (0.0116 M) where R is $-C(Me)_2CO_2Et$ (○) or $-C(Me)_2Ph$ (□). Figures adapted from ref 33.

where $[2]_0$ is the initial RAFT agent concentration and [2], [4], and [M] are the actual concentrations of RAFT agent, macro-RAFT agent, and monomer, respectively. This equation can be solved numerically to give estimates of C_{tr} *and* C_{-tr}.[32,34] However, most reported values for C_{tr} appearing in the literature are apparent transfer coefficients (C_{tr}^{app}) being based on an assumption that C_{-tr} and $k_{-\beta}$ are zero or negligible. For the more active RAFT agents, values of C_{tr}^{app} often underestimate C_{tr} by several orders of magnitude.[34]

■ ROLE OF THE Z GROUP

The Z group modifies both the rate of addition of propagating radicals ($P_n^{\bullet}$) to the thiocarbonyl of **2** and **4** and the rate of fragmentation of the intermediate radicals **3** and **5**. The rate constant k_{add} can be "adjusted" over some 5 orders of magnitude through manipulation of Z.

The most reactive RAFT agents include the dithioesters and trithiocarbonates which have carbon or sulfur adjacent to the thiocarbonylthio group. RAFT agents with a lone pair on nitrogen or oxygen adjacent to the thiocarbonyl, such as the *O*-alkyl xanthates, *N,N*-dialkyldithiocarbamates, and *N*-alkyl-*N*-aryldithiocarbamates, have dramatically lower reactivity toward radical addition. Lower rate coefficients for addition are predicted by molecular orbital calculations[34,37,38] and can be qualitatively understood in terms of the importance of the zwitterionic canonical forms **7** and **9** (Figure 2). The

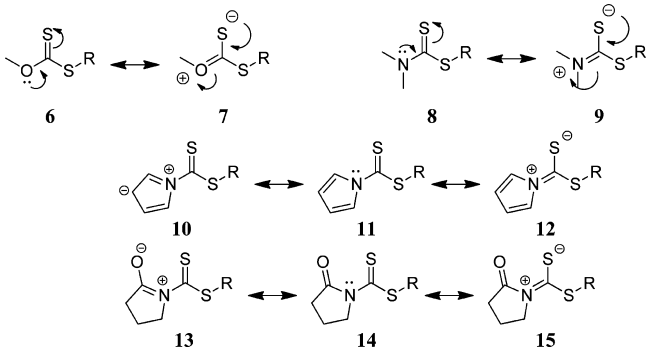


Figure 2. Canonical forms of xanthates and dithiocarbamates.

interaction between the lone pair and the C═S double bond both reduces the double-bond character of the thiocarbonyl group and stabilizes the RAFT agent (**2**) relative to the RAFT-adduct radical (**3**).[6,34,38,39] Dithiocarbamates where the nitrogen lone pair is not as available because it is part of an aromatic ring system (such as a pyrrole in **11**) or where a carbonyl (as in **14**) is α to the nitrogen lone pair (Figure 2) have reactivity similar to that of the dithioesters and trithiocarbonates.[34,40,41] In these cases, the importance of canonical forms such as **10** and **13** effectively reduces the contribution of **12** and **15**, respectively. The effectiveness of xanthates is similarly sensitive to the nature of substituents on oxygen.[39]

Monomers can be considered as belonging to one of two broad classes. The "more activated" monomers (MAMs) are those where the double bond is conjugated to an aromatic ring (e.g., styrene (St), vinylpyridine), a carbonyl group (e.g., methyl methacrylate (MMA), methyl acrylate (MA), acrylamide (AM)), or a nitrile (e.g., acrylonitrile (AN)). The "less activated" monomers (LAMs) are those where the double bond is adjacent to saturated carbon (e.g., diallyldimethylammonium chloride), an oxygen, or nitrogen lone pair (e.g., vinyl acetate (VAc) or *N*-vinylpyrrolidone (NVP)) or the heteroatom of a heteroaromatic ring (e.g., *N*-vinylcarbazole (NVC)).

Propagating radicals with a terminal more active monomer (MAM) unit are less reactive in radical addition (lower k_p, lower k_{add}), and one of the more active RAFT agents is required for good control. The poly(MAM) propagating radicals are relatively good homolytic leaving groups; therefore, retardation solely due to slow fragmentation is unlikely. The more active RAFT agents such as the dithioesters, trithiocarbonates, and aromatic dithiocarbamates allow the preparation of low-dispersity polymers from MAMs, whereas the *N*-alkyl-*N*-aryldithiocarbamates and the *O*-alkyl xanthates typically have lower transfer constants and provide poor control.

Propagating radicals with a terminal less-activated monomer (LAM) unit are highly reactive in radical addition (higher k_p, higher k_{add}). Accordingly, addition to less active transfer agents such as the *N*-alkyl-*N*-aryldithiocarbamates and the *O*-alkyl xanthates is sufficient that these RAFT agents have high transfer constants in LAM polymerization. However, the poly(LAM) propagating radicals are relatively poor homolytic leaving groups. Thus, when more active RAFT agents, such as dithioesters, are used in LAM polymerization, fragmentation is slow and inhibition or retardation is likely.

General guidelines for selection of Z are shown in Figure 3.

Irrespective of the class of RAFT agent, the transfer constant is generally enhanced by the presence of electron-withdrawing groups on Z and by the capacity of Z to stabilize an adjacent radical center.[34,37] However, these same factors generally can also increase the likelihood of side reactions (see below).

The second important role of Z is to determine the stability of the intermediate radicals **3** and **5**. When Z is aryl, the intermediate is stabilized, and the rate of intermediate radical fragmentation is slower than when connecting atom of Z is sp^3 carbon, oxygen, or nitrogen or sulfur. There currently is some unresolved controversy as to whether for dithiobenzoates the rate of fragmentation is sufficiently slow to cause retardation directly or the rate of fragmentation is simply reduced to the

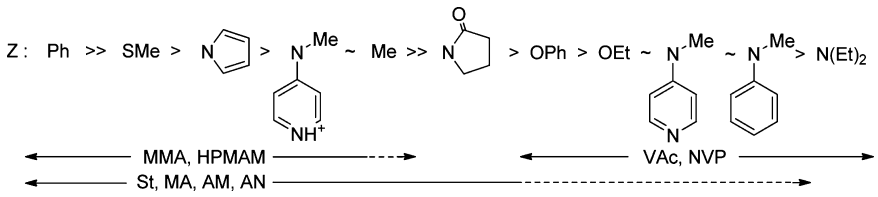


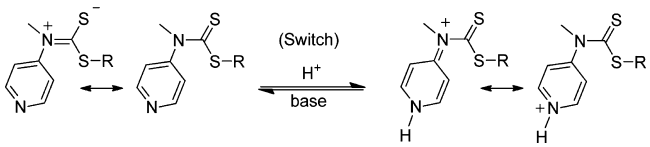
Figure 3. Guidelines for selection of the Z group of RAFT agents (ZC(═S)SR) for various polymerizations. Addition rates decrease and fragmentation rates increase from left to right. A dashed line indicates partial control (i.e., control of molar mass but poor control over dispersity or substantial retardation in the case of LAMs such as VAc or NVP). Figure adapted from that in earlier reviews.[2−5,8] HPMAM = *N*-(2-hydroxypropyl)methacrylamide.

extent that side reactions such as combination or disproportionation involving **3** or **5** are more likely. A full discussion on this topic is beyond the scope of this Perspective, and the reader is referred to the recent literature.[26−30] Irrespective of mechanism, it is clear that aromatic dithioesters give retardation. This is most apparent when higher RAFT agent concentrations are used (to give lower molecular weight polymers) and with faster propagating monomers (e.g., acrylates, vinyl esters).

It was proposed by Coote and colleagues that RAFT agents where Z is fluorine might perform as a "universal" RAFT agent.[42,43] In terms of the simple theories expounded above, the high electronegativity of fluorine means that canonical forms analogous to **7** and **9** in which an electron is removed from Z have little importance. The fluorine also provides little stability to the intermediate radical. However, difficulties in synthesis of the so-called F-RAFT agents have meant that such RAFT agents have not been fully tested.

We[44−47] have adopted a different strategy in developing RAFT agents with more universal applicability and have designed new class of stimuli-responsive, switchable, RAFT agents that can be switched to offer good control over polymerization of both MAMs and LAMs. Our motivation has been to provide a more direct route to poly(MAM)-*block*-poly(LAM). The *N*-(4-pyridinyl)-*N*-methyldithiocarbamates (Scheme 3) behave as other *N*-aryl-*N*-alkyldithiocarbamates

Scheme 3. *N*-(4-Pyridinyl)-*N*-methyldithiocarbamate Switchable RAFT Agents



and are effective in controlling the polymerization of LAMs but have relatively low transfer constants when used in MAM polymerization. However, in the presence of a strong protic or Lewis acid, the switched form of the RAFT agent provides excellent control over the polymerization of MAMs.[44]

Very recently, the *N*-aryl-*N*-(4-pyridinyl)dithiocarbamates have been evaluated in polymerizations of MA, NVC, and VAc. These RAFT agents appear more effective (dispersities are lower) than the analogous *N*-methyl-*N*-(4-pyridinium)-dithiocarbamates (Scheme 3) with LAMs in the unswitched (neutral) form and more active with MAMs in the switched (protonated) form.[48]

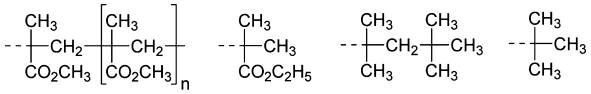
A final consideration is that Z should not cause any side reactions. With xanthates (**2**, Z = OR′) it is important that R′ be a poor homolytic leaving group. Otherwise, fragmentation with loss of R′ (irreversible chain transfer) will compete with the desired RAFT process. This requires that R′ be primary alkyl or aryl.[49,50] Similar design considerations apply in the case of unsymmetrical trithiocarbonates (**2**, Z = SR′).[51]

When Z is strongly electron-withdrawing, the thiocarbonyl group may undergo a direct reaction with monomers. Thus, RAFT agents where Z is alkylsulfonyl or phenylsulfonyl group Z = $PhSO_2$− undergo direct reaction with (meth)acrylate monomers (BA, MA, tBA, and MMA) under polymerization conditions with consumption of the thiocarbonylthio group and ultimately little control over the polymerization.[52] A hetero-Diels−Alder mechanism was suggested. Good control was achieved only with isobornyl acrylate where the side reactions with monomer were suppressed by the bulky ester substituent.

The presence of electron-withdrawing groups on Z, which lead to higher transfer coefficients, increases the likelihood of side reactions such as hydrolysis or aminolysis[2,53] and participation in cycloaddition reactions[54] such as the hetero-Diels−Alder reaction with diene monomers[52] and 1,3-dipolar cycloaddition.[55] This is an important consideration in some RAFT agent syntheses (e.g., method J, below), can be critical to the choice of RAFT agent for specific polymerization conditions (e.g., in aqueous media or in emulsion polymerization) and determines the ease of end-group transformation processes that may be required post-RAFT polymerization.

■ ROLE OF THE R GROUP

The nature of R determines the partition coefficient ϕ (eq 3). For optimal control of a polymerization, the R group of the RAFT agent (**2**, ZC(═S)SR) must be a good homolytic leaving group with respect to $P_n^{\bullet}$, such that the intermediate **3**,



16 **17** **18** **19**

formed by addition of $P_n^{\bullet}$ to **2**, both fragments rapidly and partitions in favor of **4** and $R^{\bullet}$. The expelled radical ($R^{\bullet}$) must also be able to reinitiate polymerization efficiently (i.e., $k_{i,R} > k_p$); otherwise, retardation is likely.[45] Radical stability is important in determining fragmentation rates. Experimental findings that the transfer coefficient and the value of ϕ increase in the series primary < secondary < tertiary and with the introduction of substituents which are capable of delocalizing the radical center are consistent with this view. However, other factors are of equal or greater significance.

It is not sufficient for R to be a monomeric analogue of the propagating radical because penultimate unit effects are substantial, particularly when R is tertiary. RAFT agents with R = 2-ethoxycarbonyl-2-propyl (**17**), which can be considered as a monomeric model for a methacrylate chain (**16**), provide only poor control over the polymerization of MMA and other methacrylates because R is a poor homolytic leaving group with

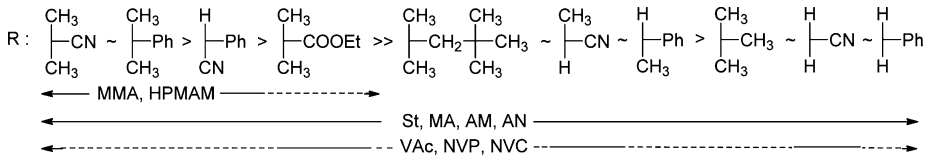
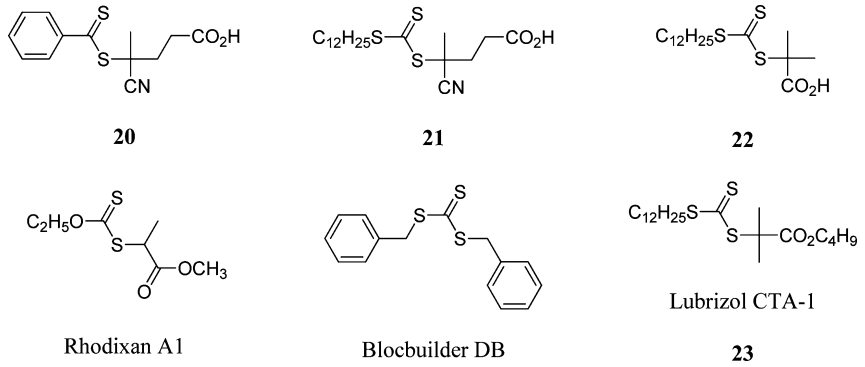
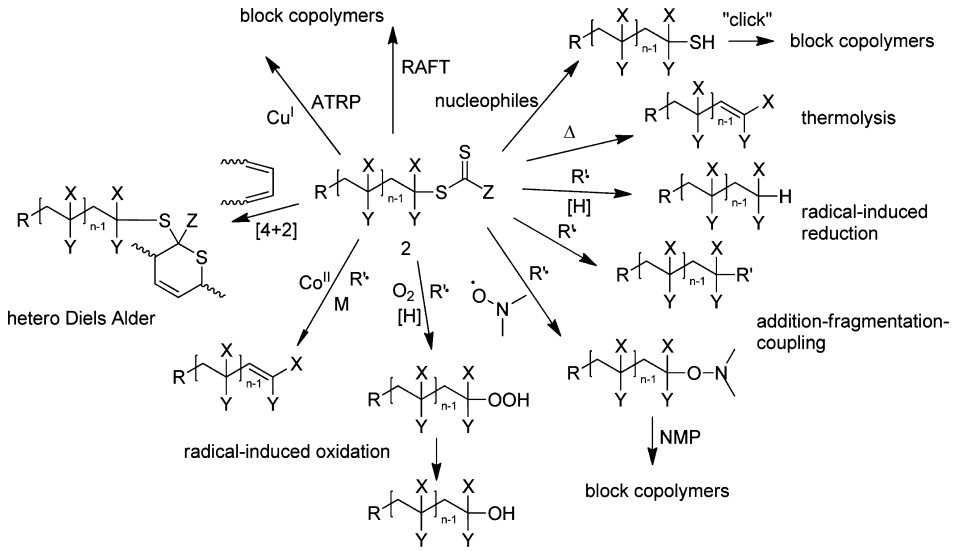


Figure 4. Guidelines for selection of the R group of RAFT agents (ZC(═S)SR) for various polymerizations. Transfer coefficients decrease from left to right. Fragmentation rates also decrease from left to right. A dashed line indicates partial control (i.e., control of molar mass but poor control over dispersity or substantial retardation in the case of VAc, NVC, or NVP). Figure adapted from that in earlier reviews.[2−5,8]



Scheme 4. Processes for RAFT End-Group Transformation (R• = Radical, [H] = Hydrogen Atom Donor, M = Monomer) [Reproduced with Permission from Ref 67. Copyright 2011 Society of Chemical Industry]



respect to the PMMA propagating radical.[33,56] For similar reasons, RAFT agent with R = *t*-butyl (**19**) is poor with respect to RAFT agent with R = *t*-octyl (**18**).[33] These differences in RAFT agent activity are attributed to steric factors.

Polar effects are also extremely important in determining the partition coefficient ϕ. Electron-withdrawing groups on R both decrease rates of addition to the thiocarbonyl group and increase rates of fragmentation. The relatively high transfer constants of cyanoalkyl RAFT agents (Figure 4) vs similar benzylic RAFT agents is attributed to the influence of polar factors.

Thus, control over the polymerization of methacrylates and methacrylamides (and other 1,1-disubsituted monomers which result in a tertiary $P_n^{\bullet}$) usually requires that R to be tertiary (e.g., 2-cyano-2-propyl or cumyl)[33] or secondary aralkyl (e.g., α-cyanobenzyl).[57,58] However, polymerization of monomers with high propagation rate constants (k_p) such as acrylates, acrylamides, vinyl esters (e.g., vinyl acetate), and vinyl amides (e.g., *N*-vinylpyrrolidone) are best controlled with RAFT agents with primary or secondary R groups. Tertiary radicals, such as 2-cyano-2-propyl radical, are inefficient in reinitiating polymerization since $k_{i,R}$ is often lower than k_p.[33]

Guidelines to the selection of R are provided in Figure 4. The order shown is based on measurements of C_{tr}^{app} for dithiobenzoate RAFT agents in St and MMA polymerization[33] and seems general based on limited data for other classes of RAFT agent. The benzylic radicals and tertiary alkyl radicals add to most LAMs very slowly (with reference to k_p) and an inhibition period is often observed with these R groups.[59]

These considerations discussed above with respect to selection of R are also important in designing the synthesis of block copolymers by RAFT polymerization. In the synthesis of a block copolymer comprising segments of a 1,1-disubstuted monomer and a monosubstituted monomer the block comprising the 1,1-disubstuted monomer should be prepared first.[33] Similarly, in synthesizing a poly(MAM)-*block*-poly-(LAM), using switchable RAFT,[44,45] the poly(MAM) block should be made first because poly(LAM) propagating radicals are relatively poor homolytic leaving groups, however, poly(MAM) propagating radicals are slow to reinitiate LAM polymerization.

■ COMMERCIAL AVAILABILITY OF RAFT AGENTS

A range of RAFT agents including the carboxylic acid functional RAFT agents **20**−**22** are now commercially available in research quantities.[60,61] The industrial scale-up has been announced of the xanthate, Rhodixan-A1, by Rhodia[62] and trithiocarbonates, Blocbuilder DB and CTA-1, by Arkema[63] and Lubrizol,[64] respectively.

■ RAFT AGENT FUNCTIONALITY

The thiocarbonylthio group is generally robust, and the RAFT process is tolerant of a wide range of unprotected functionality on Z or R, which includes hydroxy, carboxylic acid, sulfonic acid, tertiary amine, and primary, secondary, tertiary, and quaternary ammonium.

A key feature of the RAFT process is that the thiocarbonylthio groups, present in the initial RAFT agent (**2**), are retained in the polymeric product (i.e., the polymeric products of the process are also RAFT agents). They possess R group of the initial RAFT agent at one end and Z−C(═S)− group at the other. In designing functional RAFT agents and polymer architectures, it will normally be preferable to introduce that functionality through substituents on R. Any groups on Z may be lost if the thiocarbonylthio group is degraded or removed.

The Z−C(═S)− may also be transformed post-RAFT polymerization and a wide variety of processes for effecting this have been described. Recent reviews on thiocarbonylthio end-group transformation/removal include those by Willcock and O'Reilly,[65] Moad et al.,[66,67] and Barner and Perrier.[68] Thiocarbonylthio groups undergo reaction with nucleophiles and ionic reducing agents (e.g., amines, hydroxide, borohydride) to provide thiols. Thermolysis[69−72] and radical-induced reactions (e.g., addition−fragmentation transfer,[73] addition−fragmentation coupling[74,75]) are also widely used. A summary of these processes is provided in Scheme 4.[67]

The choice of Z can dictate the applicability of these processes. In general, the RAFT agents with the highest transfer constants are also most reactive in end-group transformation/removal processes. It is also important to note that while the thiocarbonylthio end group is robust, degradation can occur. For example, polymers prepared from methyl trithiocarbonates become odorous after a short period of storage due the liberation of trace amounts of methanethiol. The design of Z determines the nature of byproducts from thiocarbonylthio transformation (or unintended end-group degradation) and can also facilitate their removal.[76] Thus, trithiocarbonates should preferably be derived from a nonvolatile thiol.

■ RAFT AGENT SYNTHESIS

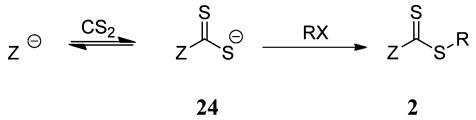
Since the advent of the RAFT process, the preparation of a vast suite of RAFT agents has been described. In this section we review the various methods for preparing RAFT agents. For each method we highlight some perceived advantages and possible drawbacks and indicate potential (or actual) side reactions. Synthetic routes to thiocarbonylthio compounds substantially predate the discovery of RAFT (and MADIX), with initial reports dating as far back as the early 1900s.[77−79] There are also a number of existing reviews on the synthesis of dithioesters and other thiocarbonyl compounds in a non-RAFT context.[80−83] The examples described in these publications often lack the specific nuances of chemical structure that underpin the requirements for an effective RAFT agent (e.g., good homolytic leaving groups). Nonetheless, the synthetic techniques described are, for the most part, applicable to the preparation of RAFT agents. This Perspective has been illustrated by examples within a RAFT context. Methods for RAFT agent synthesis are also briefly covered in some of our previous reviews of the RAFT process.[2,3,5,84]

Eight basic methods for RAFT agent synthesis (and the class of thiocarbonylthio compound that the method has generally been applied to) are: (A) reaction of a carbodithioate salt with an alkylating agent (dithioesters, trithiocarbonates, xanthates, dithiocarbamates); (B) thioacylation reactions (trithiocarbonates, xanthates, dithiocarbamates); (C) thiation of a carboxylic acid or ester (dithioesters); (D) the ketoform reaction (trithiocarbonates, xanthates, dithiocarbamates); (E) thiol exchange (dithioesters, trithiocarbonates); (F) addition of a dithioic acid across an olefinic double bond (dithioesters); (G) radical substitution of a bis(thioacyl) disulfide (dithioesters, trithiocarbonates, xanthates, dithiocarbamates); (H) radical-induced R-group exchange (dithioesters, trithiocarbonates).

The thiocarbonylthio group is moderately robust such that a wide variety of chemistries can be performed in its presence. Thus, we will also consider methods for making RAFT agents by modification of existing RAFT agents or other thiocarbonylthio compounds. This includes the synthesis of macro-RAFT agents by end-group modification of non-RAFT polymers such as biopolymers: (I) esterification of a carboxy functional RAFT agent; (J) active ester−amine reaction; (K) azide−alkyne 1,3-dipolar cycloaddition; (L) thiol reaction (Michael reaction, disulfide formation); (M) RAFT single unit monomer insertion.

A. Reaction of a Carbodithioate Salt with an Alkylating Agent. The preparation of thiocarbonylthio compounds **2** is most commonly and simply achieved by reaction of a carbodithioate salt **24** with an alkylating agent (RX) (Scheme 5). Variants of this technique are ubiquitous in the literature pertaining to RAFT agent synthesis. The method is applicable to all forms of RAFT agents (i.e., dithioesters, trithiocarbonates, xanthates, and dithiocarbamates) but is mainly used for preparation of RAFT agents containing primary and secondary R groups. RAFT agents with tertiary R are more cumbersome to prepare via this method as nucleophilic substitution on tertiary halides is relatively slow, which results in lower yields of the desired product, requires extended reaction times (e.g., Tables 1 and 2), and elimination is a potential complication.[85] Nonetheless, acceptable yields (70−80%) have been reported for some tertiary trithiocarbonates.[86−90]

Scheme 5. Preparation of RAFT Agents Using Carbodithioate Salts

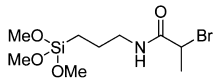
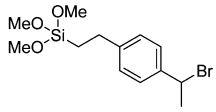
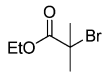


Alkyl and aryl carbodithioates as dithioester precursors are typically produced from the reaction of a Grignard reagent with carbon disulfide (some examples are provided in Table 1).[91] However, sodium salts and trialkylammonium salts have also been used.[85] Alkyllithiums are reported to provide lower yields.[83] In the example shown in Scheme 6, phenylmagnesium bromide **25** is reacted with carbon disulfide in dry THF at 40 °C to give the phenyldithioate salt **26** which, upon reaction with benzyl bromide at 50 °C for 1 h, produces benzyl dithiobenzoate **27** in 62% yield.[33]

Aryldithioate salts can also be accessed by oxidative sulfuration of benzylic halides or similar species, for example, the reaction of benzyl chloride with sodium methoxide and elemental sulfur[33,95,96] (Scheme 7, left), or by the reaction of (trichloromethyl)benzene with potassium sulfide (Scheme 7, right).[97]

For the synthesis of asymmetric trithiocarbonates (Z = *S*-alkyl or *S*-aryl, Z ≠ R), xanthates (Z = *O*-alkyl or *O*-aryl), and dithiocarbamates (Z = *N*,*N*-dialkyl, *N*-alkyl-*N*-aryl, or *N*,*N*-diaryl), carbodithioate formation usually involves reaction of the precursor Z−H (i.e., the corresponding thiol, alcohol, or amine, respectively) with carbon disulfide, in the presence of a

Table 1. Synthesis of Dithioesters from Grignard Reagents

Grignard reagent	halide	conditions[a]	yield (%)	ref
Ph		THF, 50 °C, 1h	62	33
Ph		THF, RT, 2 h	89	92
Ph		THF, 50 °C, 1 h	80	93
Ph		THF, RT, 3 h	76	58
Ph		THF, 80 °C[b], 24 h	60	94
Ph		THF, 80 °C[b], 60 h	42	33
Ph		THF, 50 °C, 24h	43	6

[a]Reaction conditions for reaction with carbodithioate salt. RT = room temperature. [b]Under reflux.

base. For example, the synthesis of *n*-butylphenylethyltrithiocarbonate (**28**) is obtained in quantitative yield from *n*-butanethiol, carbon disulfide, and (1-bromoethyl)benzene, with triethylamine as base (Scheme 8).[70]

As alcohols and amines are less acidic (higher pK_a) than thiols of similar structure,[98] the preparation of xanthates and dithiocarbamates generally requires use of stronger bases, such as sodium hydroxide[99−101] or sodium hydride,[39,102−104] to promote formation of the carbodithioate salt. The synthesis of the 2,2,2-trifluoroethylxanthate (**29**) is an example (Scheme 9).[39]

Some simple carbodithioates, such as potassium ethylxanthogenate (**30**) and sodium *N*,*N*-diethyldithiocarbamate (**32**), are commercially available allowing for synthesis of RAFT agents, such as the azide functional xanthate **31**[105] or benzyl dithiocarbamate (**33**),[34] as shown in Schemes 10 and 11, respectively.

A recent article by Skey and O'Reilly[87] further illustrated the utility of this technique for RAFT agent synthesis, making use of non-nucleophilic inorganic bases such as potassium phosphate, for preparation of trithiocarbonates (see Table 2 and entry 1, Table 3) and aromatic dithiocarbamates (see entry 2, Table 3) or cesium carbonate for synthesis of xanthates and aliphatic dithiocarbamates (see entries 3−5, Table 3). Reaction conditions were varied depending on acidity of the ZH moiety and the reactivity of the alkyl halide toward nucleophilic substitution. While some examples using tertiary alkyl halides were found to give acceptable yields of the desired products (entry 3, Table 2), these required substantially longer reaction times than those which used primary or secondary halides (cf. entries 1, 2, and 4, Table 2).

Scheme 7. Preparation of Phenyldithioate Salts from Halogenated Precursors[33,95−97]

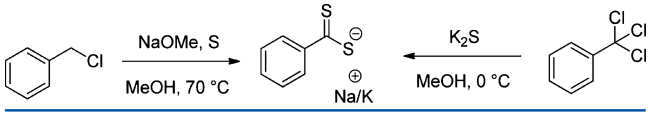


Table 2. Influence of the R-Group on the Synthesis of Dodecyltrithiocarbonates[87]

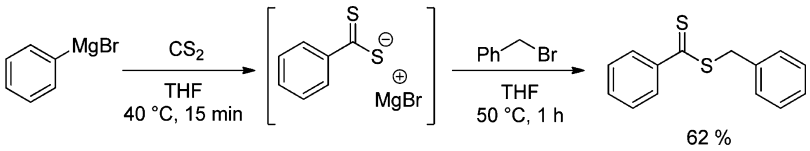
entry	thiol	halide	reaction time (h)	yield (%)[a]
1	$C_{12}H_{25}SH$	$(CH_3)CH(CN)Br$	2	71
2	$C_{12}H_{25}SH$	$(Ph)CH(CO_2Et)Br$	2	73
3	$C_{12}H_{25}SH$	$(CH_3)_2C(CO_2H)Br$	13	83
4	$C_{12}H_{25}SH$	(*N*-phthalimido)CH_2Br	0.5	59

[a]Reaction conditions: acetone, K_3PO_4, room temperature.

Dithiocarbamates prepared from aryl amines, such as **34** and **36**, require stronger bases than do aliphatic amines to achieve acceptable yields. In this case complete deprotonation of the NH bond must be facilitated, as the low nucleophilicity of the aryl-NH free base slows the reaction with CS_2 considerably.

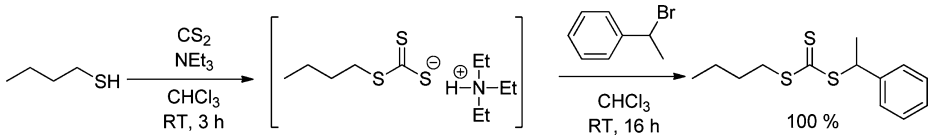
We have found that in the synthesis of the switchable dithiocarbamate RAFT agents[44,45,107] the use of butyllithium as

Scheme 6. Preparation of Benzyl Dithiobenzoate from Phenyl Grignard Reagent[33]



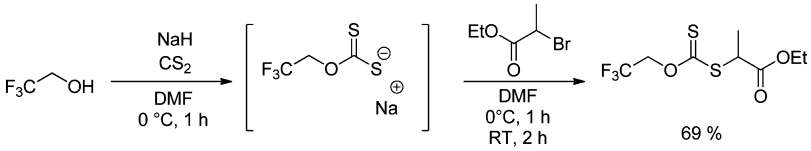
25 **26** **27**

Scheme 8. Synthesis of Butylphenylethyltrithiocarbonate[70]



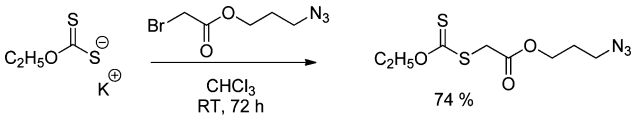
28

Scheme 9. Synthesis of 2,2,2-Trifluoroethylxanthate (29)[39]



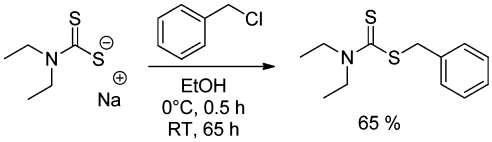
29

Scheme 10. Synthesis of Azide-Functional Xanthate 31[105]



30 **31**

Scheme 11. Synthesis of Benzyl-*N*,*N*-diethyldithiocarbamate (33)[34]



32 **33**

Table 3. Influence of the Z Group on the Synthesis of Phenylethyl and Benzyl RAFT Agents[87]

entry	ZH	halide	reaction conditions	yield (%)
1	$PhCH_2SH$	$(Ph)CH(CH_3)Br$	acetone, K_3PO_4, 4 h	91
2	imidazole	$(Ph)CH(CH_3)Br$	acetone, K_3PO_4, 4 h	78
3	CH_3CH_2OH	$(Ph)CH(CH_3)Br$	ethanol, Cs_2CO_3, 4 h	73
4	$(i\text{-Pr})_2NH$	$PhCH_2Br$	acetone, Cs_2CO_3, 10 min	61
5	$(Ph)(CH_3)NH$	$PhCH_2Br$	ethanol, Cs_2CO_3,[106] 4 h	70

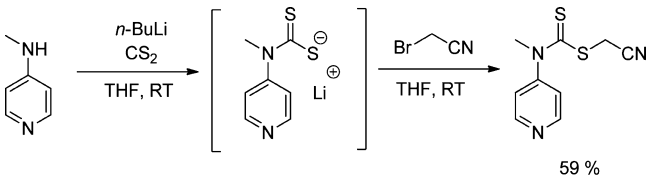
Table 4. Influence of Base Strength of Dibenzyltrithiocarbonate[112]

base (M_2CO_3)	time (h)	yield of **38** (%)
Li_2CO_3	24	<1
Na_2CO_3	24	<1
K_2CO_3	12	44
Cs_2CO_3	12	99

base can give rise to colored byproducts and when applied to the synthesis of the *N*,*N*-dipyridyldithiocarbamate (**37**, Scheme 13) gave negligible yields (0−5%) of the desired RAFT agent and a largely intractable product. Use of a strong non-nucleophilic base, potassium *tert*-butoxide (KO^tBu), in place of *n*-BuLi, resulted in a higher (45%) yield.[48]

Symmetrical trithiocarbonates, such as **38**, can be simply prepared by reaction between an alkyl halide and carbon disulfide in the presence of a base. In the case of hydroxide as base, the reaction proceeds via production of trithiocarbonate anion (CS_3^{2-}) (see Scheme 14),[108−110] which can be subsequently reacted with an alkylating agent.

Scheme 12. Synthesis of the "Switchable" Cyanomethyl Methyl(pyridin-4-yl)dithiocarbamate[44,107]



34 **35**

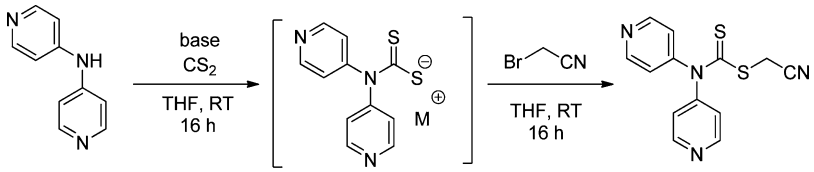
Trithiocarbonate syntheses, including those reported above, are typically carried out with a 2−3-fold excess of carbon disulfide. Aoyagi et al.[111,112] have found that it is possible to carry out the reaction with stoichiometric carbon disulfide and still obtain near-quantitative yields. However, selection of the reaction solvent and the base is critical.

A wide variety of reaction conditions for this process have been reported. Some methods employ biphasic media under phase transfer conditions,[70,113] while others use polar organic solvents, such as *N*,*N*-dimethylformamide or acetonitrile.[112] Several other variations on the general procedure appear in the literature.[114−116] Production of the trithiocarbonate anion from carbon disulfide is dependent on base strength as indicated by the yields of dibenzyl trithiocarbonate (**38**) obtained with various bases (see Scheme 15 and Table 4).[112]

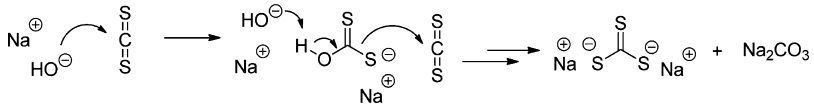
The yield of the RAFT agent prepared from carbodithioate salts is dictated on the structure and reactivity of the reagents used and the reaction conditions. The nucleophilicity of the species which must undergo reaction with carbon disulfide (Z^- in Scheme 5) is of particular importance. When Z^- is a poor nucleophile (e.g., phenoxide, thiophenoxide), this reaction can be problematic in that the equilibrium favors starting materials and other synthetic protocols (e.g., method B below) become necessary.

It should also be noted that the reactions described in this section generally make use of carbon disulfide as a reagent. Carbon disulfide is volatile (bp 46.2 °C),[117] and toxic and appropriate care should be taken.

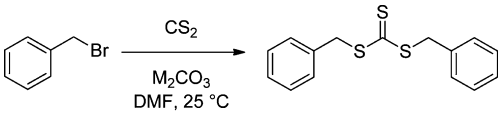
Scheme 13. Synthesis of the Cyanomethyl Di(pyridin-4-yl)dithiocarbamate[48]



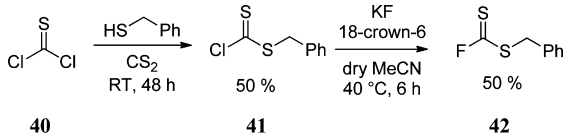
Scheme 14. Formation of Trithiocarbonate Anion from Carbon Disulfide[109]



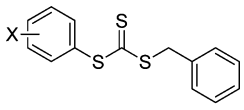
Scheme 15. Synthesis of Dibenzyltrithiocarbonate from Benzyl Bromide and Carbon Disulfide[112]



Scheme 16. Synthesis of Benzyl Fluorodithioformate, F-RAFT, from Thiophosgene[43]



B. Thioacylation Reactions. Thiophosgene (**40**) and thiophosgene equivalents, e.g., 1,1′-thiocarbonyldiimidazole (**43**), can provide an alternative route to RAFT agents for the case where Z^- does not readily form carbodithioate salts by reaction with CS_2 such that displacement chemistry (method A above) is unsuccessful or inefficient. For example, the synthesis of RAFT agents with Z = phenoxy or thiophenoxy by displacement chemistry provides, at best, only low yields because of the low nucleophilicity of the phenoxide or thiophenoxide ion and because the equilibrium in the reaction with carbon disulfide lies substantially on the side of starting materials. An alternate strategy for synthesizing such compounds is via the aryl chlorothionoformates or equivalent species. Some aryl chlorothionoformates are commercially available. Thus, *S*-benzyl (75% yield)[34] and *S*-2(-phenylethyl)-*O*-phenylxanthate[86] were prepared from the commercial phenyl chlorothionoformate. Chlorothionoformates and chlorodithionoformates can be synthesized through the reaction of the phenol or thiophenol with thiophosgene. The chlorothionoformate or chlorodithionoformate generally does not need to be isolated such that the reaction of the phenol or thiophenol (ZH) with thiophosgene, and the subsequent reaction with a thiol (R−SH), can be performed as parts of a "one-pot" process. Examples include the synthesis of *S*-benzyl-*O*-pentafluorophenylxanthate (25% yield)[34] and a series of benzylaryltrithiocarbonates (**39**) with Z = (pentachlorophenyl)thio (44% yield), (4-pyridyl)thio (37% yield), and (4-trifluoromethylphenyl)thio (23% yield).[51]

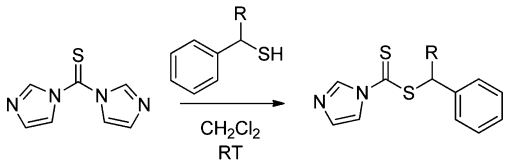


39

Barner-Kowollik and co-workers[43] used a related strategy in their preparation of the benzyl fluorodithioformate (F-RAFT, **42**) (Scheme 16). In this case, reaction of thiophosgene with benzyl mercaptan provided benzyl chlorodithioformate (**41**), which was reacted with potassium fluoride (KF) in the presence of 18-crown-6 to furnish **42** in moderate yield (50%). Other examples of this strategy include preparation of sulfonyldithioformates.[52]

It is possible to use, 1,1′-thiocarbonyldiimidazole (**43**) rather than thiophosgene in this process (Scheme 17). The preparation

Scheme 17. Preparation of Dithiocarbamates 44 and 45 from 1,1′-Thiocarbonyldiimidazole[86,118]

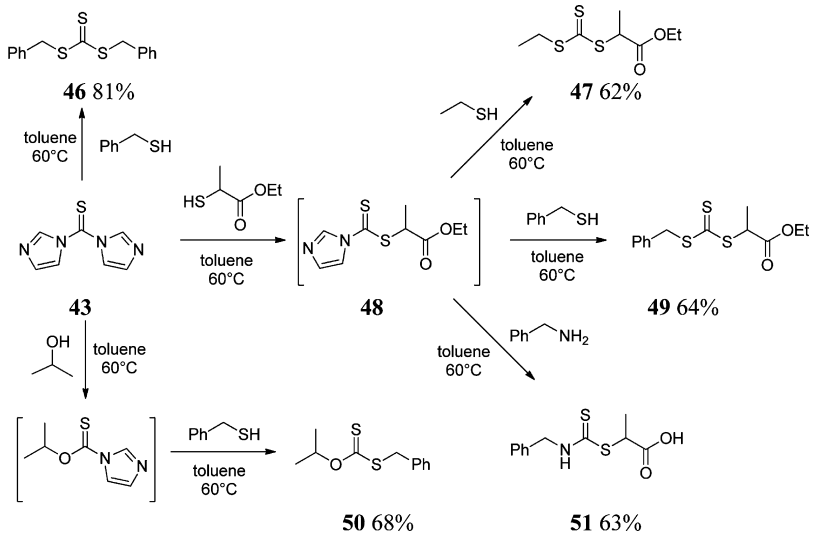


of benzyl 1-imidazolyldithiocarbamate (**44**) by reaction with benzyl mercaptan was reported in a CSIRO patent,[118] and its use as a RAFT agent was subsequently published.[40] Phenylethyl 1-imidazolyldithiocarbamate (**45**) has been prepared by the same method.[86] Bicciocchi et al.[51] showed that use of **43** allowed for a convenient route to substituted arylbenzyl trithiocarbonates **39**.

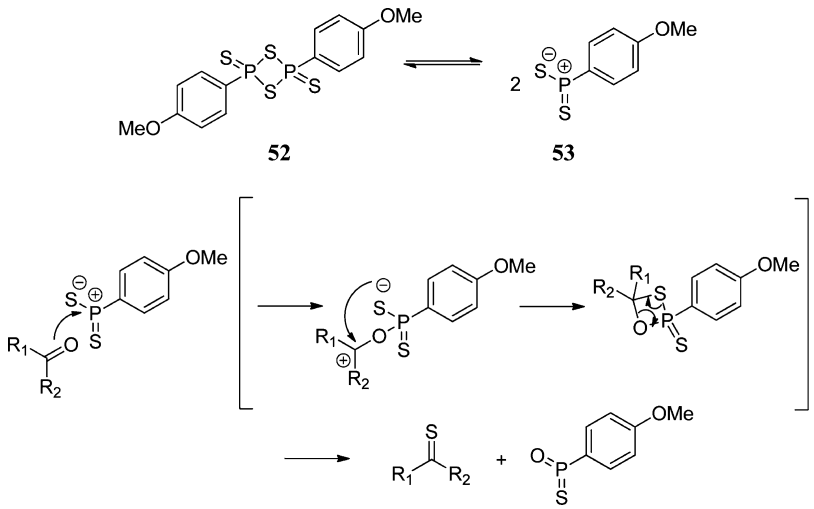
Perrier and co-workers have used this methodology for the synthesis of a range of trithiocarbonates (**46**, **47**, and **49**), dithiocarbamates (**48** and **51**), and the xanthate **50** achieving moderate to high yields (Scheme 18).[119] To use this reagent, Z must be a better nucleophile than imidazole. Thiophosgene has advantages in this context since chloride ion is a better leaving group than imidazolide anion. However, the higher reactivity and toxicity of thiophosgene make it a more difficult reagent to handle.

While the use of thiophosgene (**40**) or 1,1′-thiocarbonyldiimidazole (**43**) allows for the synthesis of RAFT agents with Z

Scheme 18. Use of 1,1′-Thiocarbonyldiimidazole for the Synthesis of Trithiocarbonates, Dithiocarbamates, and Xanthates[119]



Scheme 19. Mechanism of Thiation with Lawesson's Reagent[125]

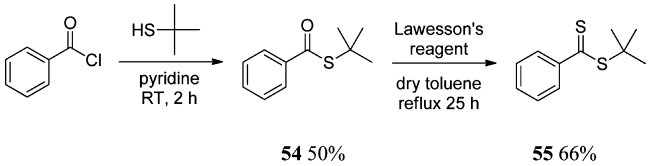


group substitution that may be difficult to prepare via alkylation of a carbodithioate salt, this technique suffers from some limitations. Some desirable R groups, such as cyanomethyl or 2-cyano-2-propyl, are not easily accessed by this method because the required thiols are commercially unavailable and difficult to handle when prepared.[120−122]

C. Thiation of Carboxylic Acids and Esters. Thiation of carbonyl compounds to form thiocarbonyl compounds can be facilitated by the use of phosphorus pentasulfide, Lawesson's reagent (LR, **52**),[123] and other tetrathiophosphates, such as Davy reagent.[124] A proposed mechanism[125] with LR involves reaction of the reactive dithiophosphine ylide **53** with the carbonyl group of the substrate and subsequent cleavage of a four-membered heterocycle as shown in Scheme 19. A comprehensive review of the rich chemistry of LR has been provided by Jesberger et al.[125]

An example of the use of LR in a RAFT context are the synthesis of phenylethyl dithiobenzoate[126] and *tert*-butyl dithiobenzoate (**55**).[33] Treatment of benzoyl chloride with *tert*-butyl mercaptan gave the thioloester **54** (50%), which was transformed to the dithioester in reasonable yield (66%) by refluxing with LR **52** in toluene (Scheme 20).[33]

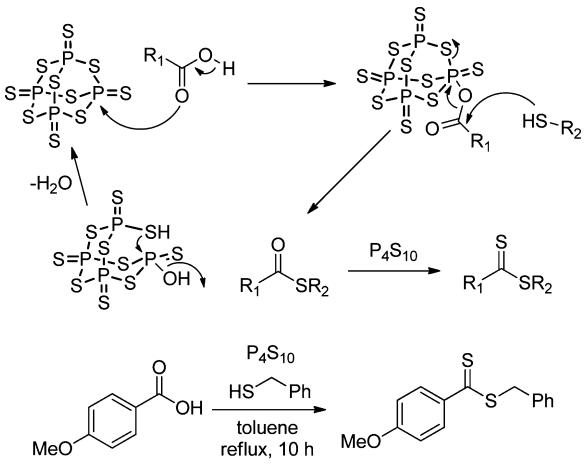
Scheme 20. Thiation of Benzoyl Thioloester with Lawesson's Reagent[33]



Sudalai et al.[127] reported use of P_4S_{10} for preparation of dithioesters from carboxylic acids and either thiols or alcohols. The reaction is suggested to proceed by formation of an activated carboxylic acid derivative by reaction with P_4S_{10}, which can be readily attacked by a thiol (which was added directly to the reaction mixture or formed *in situ* from P_4S_{10} and an alcohol), as shown in Scheme 21 (top). Thiation of the carbonyl group then proceeds similarly to that described above for **52**.

An example of such a reaction is the production of benzyl 4-methoxydithiobenzoate, **56** (see Scheme 21, bottom).

Scheme 21. Mechanism and Example of the Formation of Dithioesters from Carboxylic Acids with Thiols/Alcohols and P_4S_{10}[127]



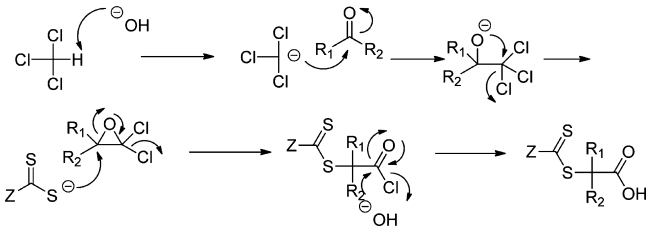
Dureault et al.[128] have prepared di- and trifunctional RAFT agents from the corresponding di- and tricarboxylic acids and benzyl mercaptan by a similar technique.

Dureault et al.[129] also formed RAFT agents *in situ* from carboxylic acids in St polymerization. In this process a phenylethyl R group is incorporated, by Markovnikov addition of the dithioic acid to St (section F).

Some drawbacks of this technique include the limited availability of some desired carbonyl and thiol/alcohol precursors. In their report, Sudalai et al.[127] indicate the occurrence of side reactions in some examples, such as Friedel–Crafts alkylation of the toluene solvent when alcohol precursors were used. This suggests that the method may not be tolerant to some functional groups. It is not clear how universal the protocol is as to date only dithioesters have been prepared using this technique.

D. RAFT Agents Prepared via the Ketoform Reaction. Lai and co-workers have adapted the ketoform reaction[130] as a method to prepare trithiocarbonates,[131] dithiocarbamates, and xanthates[132] containing tertiary carboxyl groups. The proposed mechanism (Scheme 22) involves ring-opening of a

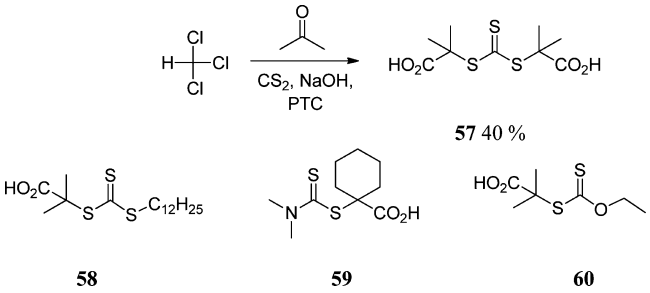
Scheme 22. Proposed Mechanism of the Ketoform Reaction[130,131]



dichloroepoxide intermediate with a nucleophile,[131] in this case a carbodithioate salt, as a key step. Subsequent hydrolysis of the acyl chloride produced provides the acid functional RAFT agent.

Reaction of carbon disulfide, chloroform, a ketone (acetone in most examples, although others have been used), and sodium hydroxide in the presence of a phase transfer catalyst (PTC) furnishes symmetrical trithiocarbonates (e.g., Scheme 23).

Scheme 23. Synthesis of Tertiary Carboxyl-Functional RAFT Agents via the Ketoform Reaction

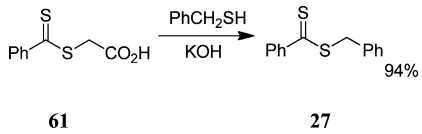


Inclusion of a thiol, amine, or alcohol in the reaction mixture allows asymmetric trithiocarbonates, dithiocarbamates, or xanthates to be prepared respectively (see lower portion of Scheme 23 for examples). The technique has been successfully implemented by a number of research groups[133−139] with reported yields varying from ∼35% to quantitative. Lubrizol has successfully developed the process to produce the trithiocarbonate **23** in multiton quantities.[140]

The advantage of this technique is that it allows synthesis of a variety of RAFT agents containing tertiary carboxylic acid R groups in a single preparative step. As the reaction medium is biphasic (part aqueous), carbodithioate salts derived from intermediates not compatible with water (e.g., those derived from anilines)[132] need to be prepared in a separate step. Attention must be paid to stoichiometry and reagent volatility and toxicity (acetone, chloroform, carbon disulfide).

E. Thiol Exchange. Thioglycolic acid-based dithioesters have been used as precursors to other RAFT agents.[141−143] They undergo facile reaction with other thiols to provide new dithioesters. For example, the reaction of **61** with benzylmercaptan provides benzyl dithiobenzoate (**27**) in high yield (Scheme 24).[144] In a RAFT context, the scope of the process is limited by the ready availability of thiols of appropriate structure.

Scheme 24. RAFT Agent Synthesis by Thiol Exchange[144]

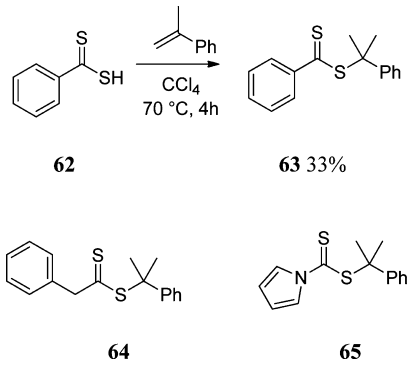


When the reaction can be carried out in aqueous solution, and product solubility is such that the desired dithioester separates from the reaction medium, the equilibrium is shifted to the product side, resulting in complete transesterification. For reactions carried out in organic solvents, treatment of the reaction mixture with aqueous alkali preferentially removes the thioglycolic acid byproduct, again resulting in near-quantitative conversion to the required dithioester.[141]

F. Addition of a Dithioic Acid across an Olefinic Double Bond. The reaction of a dithioic acid with a variety of electron-rich alkenes was reported to provide quantitative yields of the Markovnikov addition product.[145] The olefins used included St, α-methylstyrene, VAc, vinyl ethers, and vinyl

sulfones. We found the reaction of dithiobenzoic acid (**62**) with α-methylstyrene in CCl_4 provided cumyl dithiobenzoate (**63**) in relatively low yield (33%) after purification (Scheme 25).[6]

Scheme 25. Synthesis of Cumyl Dithiobenzoate by Markovnikov Addition of a Dithiobenzoic Acid to α-Methylstyrene[6]



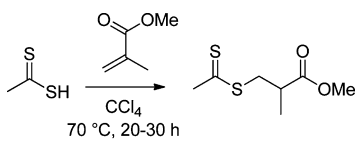
The similar methodology has also been applied to prepare cumyl dithioacetate (45% yield),[34] cumyl phenyldithioacetate, **64** (36% yield),[146] cumyl pyrrolocarbodithioate, **65** (28% yield),[147] 2-phenylethyl dithiobenzoate (43% yield),[33] and 2,4,4-trimethylpent-2-yl dithiobenzoate (32% yield).[33]

Kanagasabapathy et al.[126] have published on the regioselectivity of addition of thiols and thiobenzoic acids to St. In contrast to the behavior seen with dithioic acids, in the absence of an acid catalyst these reagents react with St and other substrates to give predominantly (>94% for St) anti-Markovnikov addition. Only in the presence of a Lewis acid (montmorillonite K10) was the Markovnikov product obtained with a high degree of regioslelectivity. The thionoester formed by reaction of thiobenzoic acid to St was converted to the dithioester phenylethyl dithiobenzoate through treatment with Lawesson's reagent (method C).[126]

Much has been published on retardation mechanisms in RAFT polymerization, and much of this literature relates to polymerizations with cumyl dithiobenzoate.[26] Monteiro and co-workers[148] have suggested that the purity of cumyl dithiobenzoate as formed by this process and, in particular, the presence of dithiobenzoic acid, may be responsible for the retardation and indicated the inadequacy of silica chromatography for purification. In our work[6] we used chromatography on neutral alumina to provide a product which retained effectiveness for >2 years when stored in the refrigerator.

With electron-poor olefins, such as (meth)acrylates, (meth)acrylamides, or (meth)acrylonitrile (e.g., MMA; Scheme 26), the Michael addition (or anti-Markovnikov product) predominates.[145] The primary alkyl radicals are poor homolytic leaving groups. Consequently, compounds such as **66** do not have utility as RAFT agents.

Scheme 26. Michael Addition of Dithioacetic Acid to Methyl Methacrylate[145]



66

G. Reaction of Radicals with Bis(thioacyl) Disulfides. Radical-induced decomposition of bis(thioacyl) disulfides, such as **67**, by radicals derived from azo-initiators is arguably the most effective technique for the synthesis of functional tertiary RAFT agents (see Scheme 27 and Table 5).[66,149,150] The method has been applied to each of the most common forms of RAFT agent (dithioesters, trithiocarbonates, xanthates, and dithiocarbamates). The usual radical source is an excess of an azo-compound initiator,[66,149,150] and the main contaminants in the product are the byproducts of azo-compound decomposition (e.g., tetramethylsuccinonitrile when azobis(isobutyronitrile) (AIBN) is used).

In our work, the bisthioacyl disulfide has usually been prepared by oxidation of the carbodithioate salt with molecular iodine which generally proceeds in quantitative yield.[66,73] Alternate methods include oxidation of a carbodithioate salt with potassium ferricyanide and the reaction of carbodithioate salt with *p*-toluenesulfonyl chloride.[151]

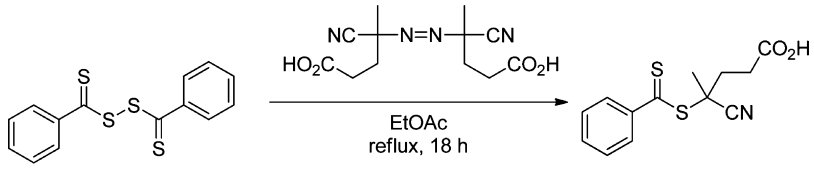
A related process has been reported by Wager et al.[154] and Kwak et al.[155] in which an ATRP initiator (a secondary or tertiary alkyl halide and a copper catalyst) is used as the radical source. The latter group termed the process atom transfer radical addition–fragmentation (ATRAF).[155] For dithioesters reported yields were generally high (82–92%), but lower yields were observed for dithiocarbamate (65%) and xanthate (43%) examples.[155] Wager et al.[154] also demonstrated the preparation of macro-RAFT agents from ATRP-synthesized polymers using this strategy.

H. Radical-Induced R-Group Exchange. A useful technique for the preparation of RAFT agents involves R-group exchange by placing a RAFT agent in the presence of a radical source such as a thermal initiator. For the process to be effective, the R group of the precursor RAFT agent should be a good homolytic leaving group with respect to that of the desired RAFT agent.[156] Thus, tertiary R groups cannot be efficiently introduced through reaction of tertiary radicals with RAFT agents possessing primary or secondary R groups.

An early example was the preparation the 2-cyano-2-propyl dithiobenzoate from cumyl dithiobenzoate by heating the latter in the presence of AIBN as a source of 2-cyano-2-propyl radicals.[33] The same process was used to prepare 2-cyano-2-propyl dithioacetate from cumyl dithioacetate.[34] Alberti et al.[156] further developed this methodology when they employed a "RAFT agent" **69** which has a triphenylmethyl R group for the preparation of the 2-cyano-2-propyl phosphoryldithioformate (**70**, Scheme 28). The corresponding bis(thioacyl) precursor (for method G) could not be prepared.[156] The same process was used to prepare 2-cyano-2-propyl dithiobenzoate[156] and 2-cyano-2-propylpyridin-4-ylcarbodithioate.[157] The driving force of the reaction is the production of a stable triphenylmethyl radical.

The use of macro-RAFT agents in this process has been developed as a method for end-group removal and for regenerating a (low molecular weight) RAFT agent.[74] Thus, PMMA with dithiobenzoate ends was heated with a 20-fold excess of AIBN to both cleave the end group (the PMMA propagating radicals undergo radical–radical termination) and produce 2-cyano-2-propyl dithiobenzoate.[74] Many additional examples of the use of this method for end-group removal have been reported.[6] While the method was stated to have general applicability to RAFT-synthesized polymers,[74] it appears inefficient, often giving incomplete end-group removal, when

Scheme 27. Synthesis of a Tertiary RAFT Agent from a Bis(thioacyl) Disulfide[149]



67 **68** 68%

Table 5. Synthesis of RAFT Agents by Reaction of Azo-Compounds R−N=N−R with Bis(thioacyl) Disulfides [ZC(=S)S]$_2$

Z	R	reaction conditions[a]	yield (%)[b]	ref
Ph	$C(CH_3)(CN)$ $CH_2CH_2CO_2H$	A	68	149, 152
Ph	$C(CH_3)(CN)$ $CH_2CH_2CO_2C_6F_5$	A	87	153
Ph	$C(CH_3)(CN)$ $CH_2CH_2CH_2OH$	B	46	33, 149
Ph	$C(CH_3)_2(CN)$	A	69	149, 150
Ph	$C(CH_3)_2(CO_2CH_3)$	D	65	150
CH_3S	$C(CH_3)_2(CN)$	C	47	34, 149
$CH_3(CH_2)_{11}S$	$C(CH_3)(CN)$ $CH_2CH_2CO_2H$	A	87	66
$CH_3(CH_2)_{11}S$	$C(CH_3)_2(CN)$	A	60	73
C_2H_5O	$C(CH_3)_2(CN)$	A	94	149, 150
C_2H_5O	$C(CH_3)(CN)$ $CH_2CH_2CO_2H$	E	77	150
(*N*-pyrrole)	$C(CH_3)_2(CN)$	B	61	34, 149
$C(CH_3)_2N$	$C(CH_3)_2(CN)$	C	93	149, 150

[a]Reaction conditions: (A) reflux, ethyl acetate, 18 h; (B) degas, ethyl acetate, 70 °C, 24 h; (C) reflux, benzene, 24 h; (D) reflux, cyclohexane; (E) reflux, dioxane/cyclohexane. [b]Isolated yield after purification.

applied to styrenics and acrylates[75] and does not work with poly(LAM).[67]

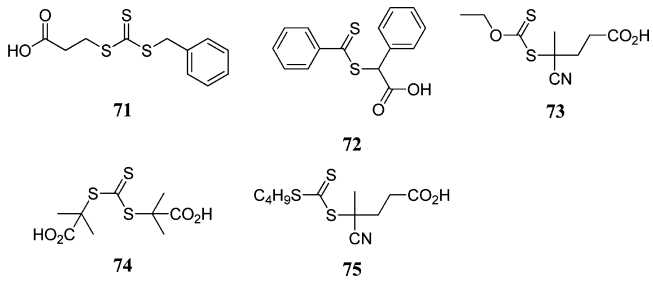
Macro-RAFT Agent Synthesis. In this section we address the synthesis of macro-RAFT agents or functional RAFT agents by modification of other RAFT agents. In particular, we consider the synthesis of macro-RAFT agents based on oligomers or polymers that are not formed by RAFT polymerization. Two areas that have recently attracted much attention are the production of biopolymer (peptide, protein, siRNA, polysaccharide) conjugates[158−163] for biomedical applications and block copolymers containing fully conjugated segments[164] (e.g., a poly(3-hexylthiophene segment[165]) for optoelectronic applications.

There are two basic strategies for forming such structures that are relevant to this section. The first involves formation of a biopolymer based macro-RAFT agents which is then used in RAFT polymerization. The second involves forming a polymer by RAFT polymerization that possesses functionality to facilitate conjugation post-RAFT polymerization. It should be noted in this context that a number of processes for forming block copolymers post-RAFT polymerization are based on direct modification of the thiocarbonylthio end-group. Processes for RAFT end-group transformation are mentioned briefly below but are more fully detailed in our recent review of that topic.[67]

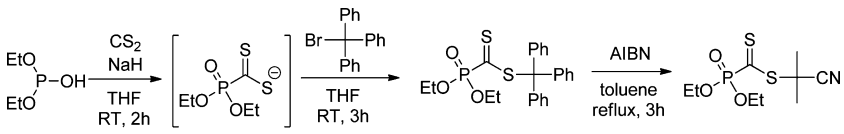
I. Esterification or Amidation of Acid Functional RAFT Agent. One of the most common methods to introduce new functionality into a RAFT agent or form a macro-RAFT agent is to react a hydroxy or amino functional substrate with a carboxy functional RAFT agent. Standard synthetic protocols for the synthesis of esters and amides, such as the acid chloride route[166,167] or carbodiimide coupling reaction,[168−172] are compatible with the thiocarbonylthio group and generally give very high yields. Acid functional RAFT agents used in this context include **20**, **22**, and **71**−**75**. Some representative examples of process are provided in Tables 6 and 7, and many additional examples can be found in our previous reviews.[2,3,5]

J. Active Ester−Amine Reaction. The active ester−amine reaction has been used to prepare of functional RAFT agents from substrates bearing amine functionality (Scheme 29). A variety of RAFT agents bearing active-ester functionality have been prepared, including pentafluorophenyl ester (**76**),[153] *N*-hydroxylsuccimidyl ester (**77**),[174,175,186] and 2-mercaptothiazoline ester (**78**).[176,187] Some examples appear in Table 6. The strategy has been widely used for biopolymer conjugation. The sensitivity of the thiocarbonylthio group to aminolysis means that care must be taken in selecting reaction conditions and stoichiometry (the amine should not be used in excess).[67]

A specific example is the synthesis of the biotin macro-RAFT agent **80** from RAFT agent **79** (Scheme 30).[186]

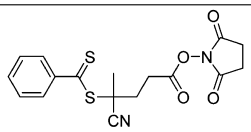
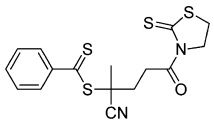
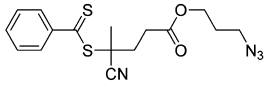
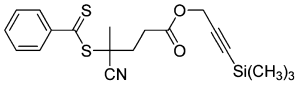
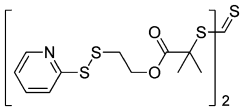
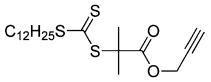
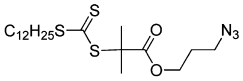
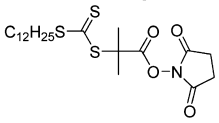
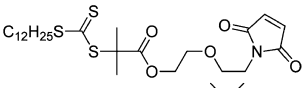
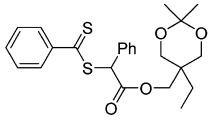


Scheme 28. Synthesis of Tertiary RAFT Agents by Radical-Induced R-Group Exchange



69 81% **70** 80%

Table 6. Examples of Functional RAFT Agents Formed from Acid Functional RAFT Agents

Acid RAFT agent	Functional RAFT agent	Reagents[a]	Reported Yield (%)	ref
20		DCC DCC/DMAP	[b] 90	173,174 175
20		DCC/DMAP	83	176
20		EDC/HOBT	34	177
20		EDC/DMAP	88	105
74		$SOCl_2$[c]	56	178
22		EDC/DMAP	[b]	179
22		oxalyl chloride[c] DCC/DMAP	90 86	177 180
22		DCC	95	181
22		oxalyl chloride[c]	58	182
72		DCC/DPTS	55	183

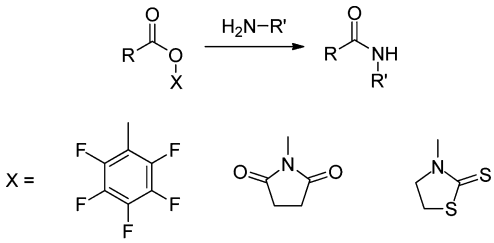
[a]DCC = *N*,*N*′-dicyclohexylcarbodiimide, DMAP = 4-(dimethylamino)pyridine, DPTS = 4-(*N*,*N*-dimethylamino)pyridinium-*p*-toluenesulfonate, EDC = *N*-(3-(dimethylamino)propyl)-*N*′-ethylcarbodiimide hydrochloride, and HOBT = 1-hydroxybenztriazole. [b]Yield not reported. [c]Process involves formation acid chloride with reagent indicated.

Table 7. Examples of Macro-RAFT Agents Formed from Acid Functional RAFT Agents

acid RAFT agent	block A[a] (end group)	reagents[b]	yield (%)	ref
22	PLA (OH end)	oxalyl chloride[c]	93	167
71	PDMS (OH end)	DCC	–[d]	168
20	PEG (OH end)	DCC/DMAP	88–100	96, 172, 184
73	PEG (OH end)	DCC	94	169
75	P3HT (OH end)	DCC/DMAP	94	185
22	peptide (NH_2 end)	DIC/NMI	100	170
20	peptide (NH_2 end)	DCC	76	171

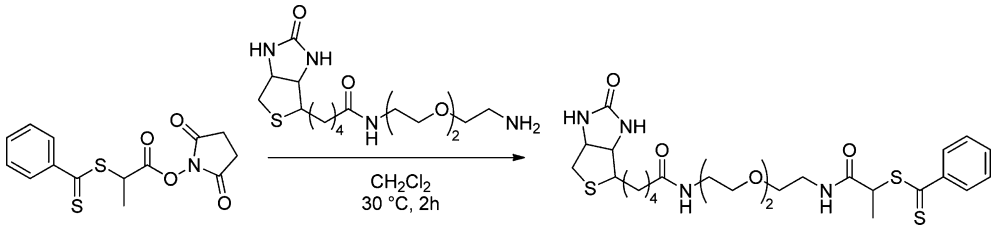
[a]P3HT = poly(3-hexylthiophene), PLA = polylactide, PDMS = poly(dimethylsiloxane), PEG = poly(ethylene glycol). [b]DCC = *N*,*N*′-dicyclohexyl carbodiimide, DPTS = 4-(dimethylamino)pyridinium-4-toluenesulfonate, DIC = *N*,*N*′-diisopropyl carbodiimide, NMI = *N*-methylimidazole. [c]Process involves formation acid chloride with reagent indicated. [d]Yield not reported.

Scheme 29. Active Ester–Amine Reaction and the Structure of Some Common Active Esters



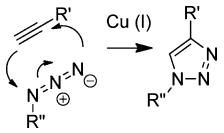
K. Azide–Alkyne 1,3-Dipolar Cycloaddition. There have been many papers on the use of RAFT polymerization in combination with copper-catalyzed azide–alkyne 1,3-dipolar cycloaddition (Scheme 31).[188–193] Many of these deal with the formation of an alkyne or azide functional polymer through the use of a RAFT agent with the corresponding functionality. A wide range of alkyne or azide functional RAFT agents have been reported (for some examples see Table 6). Ladmiral et al.[194] have pointed out that azides can undergo 1,3-dipolar cycloaddition with many common monomers (MMA, MA, *N*-isopropylacrylamide (NIPAM), and St were studied) and that this can occur under polymerization conditions. The use of

Scheme 30. Synthesis of a Biotin Functional RAFT Agent via the Active Ester−Amine Reaction[186]



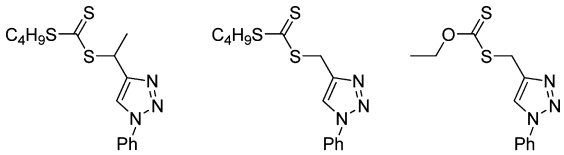
79 **80** 72 %

Scheme 31. Copper-Catalyzed Azide−Alkyne 1,3-Dipolar Cycloaddition



lower reaction temperatures during polymerization can minimize this problem. The importance of protecting alkyne functional RAFT agents (or monomers) as the trimethylsilyl derivative has been regarded as important by some authors.[105] However, in other cases reagents with unprotected alkyne functionality have been used with apparently minimal (no reported) side reactions, which is attributed to the alkyne being much less reactive toward radical addition than a (meth)acrylate double bond.

RAFT agents having a triazolinylmethyl R group have been synthesized by a copper-catalyzed 3 + 2 cycloaddition reaction between an azide and a propargyl thiocarbonylthio compound.[195] The triazolinyl trithiocarbonates **81** and **82** were effective in controlling polymerizations of St and butyl acrylate (BA) while the triazolinyl xanthate **83** was able to control polymerizations of NVP and VAc.



81 **82** **83**

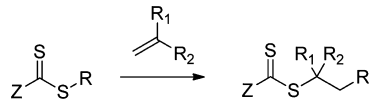
L. Thiol Reactions. Modification of RAFT agent functionality by reaction with a thiol using so-called "thiol-click" reactions,[196] such as the Michael thiol−ene reaction or disulfide formation, has been widely exploited in modification of RAFT-synthesized polymers post polymerization.[67] These reactions have not generally been considered as a suitable method for preparing RAFT agents because of the likelihood of side reactions. However, such methods have been successfully used to prepare macro-RAFT agents through conjugation to cystein functionality in bovine serum albumin (BSA).

Thus, Sumerlin and co-workers used the thio Michael reaction of a maleimide functional RAFT agent to prepare a BSA macro-RAFT agent as shown in Scheme 32.[182,197] The possibility of side reactions (e.g., thiol exchange; cf. method E above) was reduced by using the RAFT agent in very large (20×) excess over BSA thiol residues which were quantitatively converted.

Boyer et al.[198] used disulfide coupling to prepare a BSA macro-RAFT agents from a RAFT agent with pyridyl disulfide functionality (Scheme 33). Again, the process involved using the RAFT agent in very large excess over BSA thiol functionality to quantitatively convert those groups and avoid side reactions.

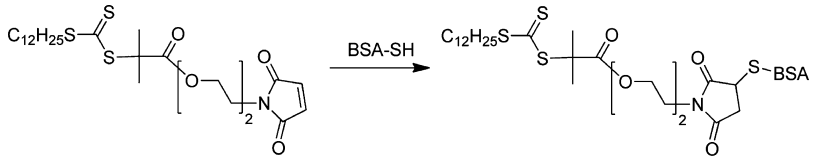
M. Single Unit Monomer Insertion. New thiocarbonylthio compounds with the general structure **85** can be prepared by the addition of a single monomer unit to a preprepared RAFT agent **84**, as illustrated in Scheme 34. Single unit monomer insertion is favored by using equimolar monomer and RAFT agent. A high C_{tr} for the RAFT agent and a high rate of addition of the radical ($R^{\bullet}$) to monomer relative to further propagation are also important. This ensures that less than one monomer unit is incorporated per activation cycle. The value of C_{tr} is determined by the relative rate of addition to the

Scheme 34. Single Unit Monomer Insertion into a Thiocarbonylthio RAFT Agent

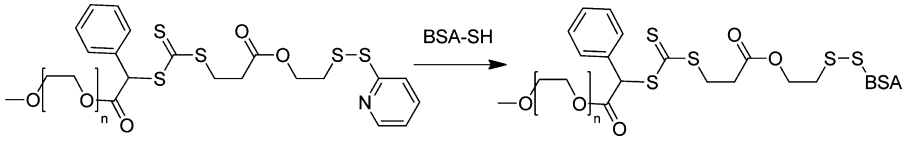


84 **85**

Scheme 32. Preparation of a RAFT Agent by the Michael Reaction (BSA = Bovine Serum Albumin)



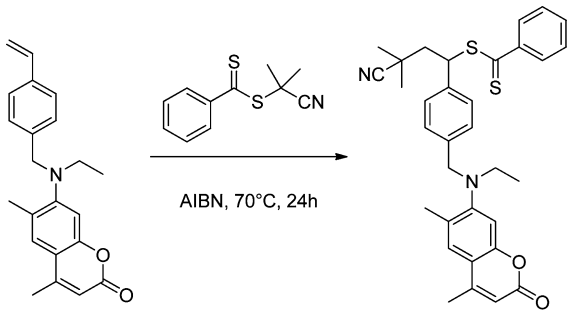
Scheme 33. Preparation of a RAFT Agent by the Disulfide Coupling (BSA = Bovine Serum Albumin)



RAFT agent vs monomer and the way the intermediate radical partitions between starting materials and products.[32,33] Thus, to ensure a high rate of addition, the RAFT agent needs to be chosen for the monomer to be inserted.[6,34] Typically this will mean use of xanthates or dithiocarbamates should be preferred for LAMs and trithiocarbonates, dithioesters, or the more active form of switchable RAFT agents[44−47] for more activated monomers (MAMs). Under these conditions "selective initialization"[199−203] occurs, and the initial RAFT agent starting material **84** is converted to the desired single unit insertion product before any significant oligomerization or polymerization occurs.

Zard and co-workers pioneered the use of this chemistry to carry out a wide range of organic transformations which involve insertion of single units of unactivated alkenes (LAMs) into xanthate RAFT agents (Z = OR).[204,205] Chen et al.[206] demonstrated that with appropriate choice of reaction conditions a similar methodology could be applied to perform single unit monomer insertion of MAMs to form macro-RAFT agents. One of the first examples reported was insertion of the St derivative **86** into 2-cyano-2-propyl dithiobenzoate to provide **87**.

Scheme 35. Preparation of a RAFT Agent by Single-Unit Monomer Insertion of Coumarin-Functionalized Styrene[206]

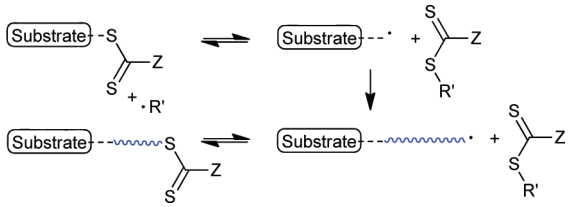


Other recent examples of single unit monomer insertion include vinylphosphonic acid,[207] St derivatives,[208−210] NIPAM,[210] and 2-vinylthiophene derivatives (including a poly(3-hexylthiophene) macromonomer).[165] While successful single unit monomer insertion into an initial low molecular weight RAFT agent can be successfully performed to provide high yields of new macro-RAFT agents, there remain some challenges in using the methodology to performing successive single unit insertions.[210] In particular, initiator-derived byproducts become an issue when R is different from the initiator-derived radical.

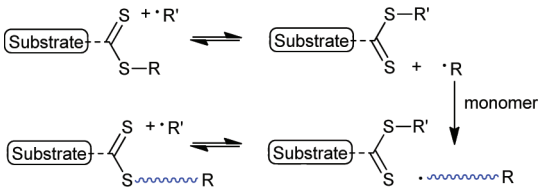
Agents for RAFT Surface Modification. A variety of chemistries have been exploited for attaching RAFT agent (or initiator) functionality to surfaces for use in "grafting from" or surface-initiated polymerization.

So-called "away from" processes, where Z is bound to the substrate (Scheme 37), have an advantage over "attached to" processes, where R is bound to the substrate (Scheme 36), in that during RAFT polymerization the propagating radicals are never directly attached to the surface. Termination involves the reaction of "free" propagating radicals in solution to produce a byproduct that can be washed away. The thiocarbonylthio functionality remains directly attached to the surface. It has been suggested that steric factors associated with attack of the propagating radical on the surface-bound RAFT functionality could become an issue, particularly at high conversions.[211] A potential disadvantage of the "away from" strategy is that any reaction which cleaves

Scheme 36. "Attached To" Process for "Grafting From" with a R-Connected RAFT Agent



Scheme 37. "Away From" Process for "Grafting From" with a Z-Connected RAFT Agent



the thiocarbonylthio groups (e.g., hydrolysis, thermolysis) also results in the loss of the graft. With the "attached to" strategy (Scheme 36) most propagating species remain attached to the surface, and the thiocarbonylthio functionality is maintained at the chain ends. Termination can result in cross-linking.

Methods used for preparing RAFT agent-functionalized surfaces have been reviewed.[212] The methods used include those used for the synthesis of other macro-RAFT agents such as esterification or amidation (method I) in the case of graphene oxide,[213] cellulose,[214] or nylon surfaces;[215] active ester-amine reaction (method J) for silica[176] and proteins;[216] azide−alkyne 1,3-dipolar cycloaddition (method K) for silica[179,217,218] and gold nanoparticles;[136,219] thio Michael reaction for proteins (method L);[182,197] single unit monomer insertion (method M) for St−divinylbenzene microspheres.[220] Other methods have also been used for specific substrates such as electrodeposition for conductive ITO or gold surfaces.[221−223]

While many processes for surface-initiated RAFT polymerization involve attaching the RAFT agent functionality to the substrate directly, another commonly used approach involves forming radicals on the surface (e.g., by irradiation or from attached initiator functionality) to initiate polymerization in the presence of a "free" RAFT agent which becomes attached to the surface during RAFT polymerization in a process analogous to radical-induced R group exchange (method H above). The mechanism is then the same as that shown in Scheme 36. The approach has been used to modify silica[224] and polypropylene[225] surfaces.

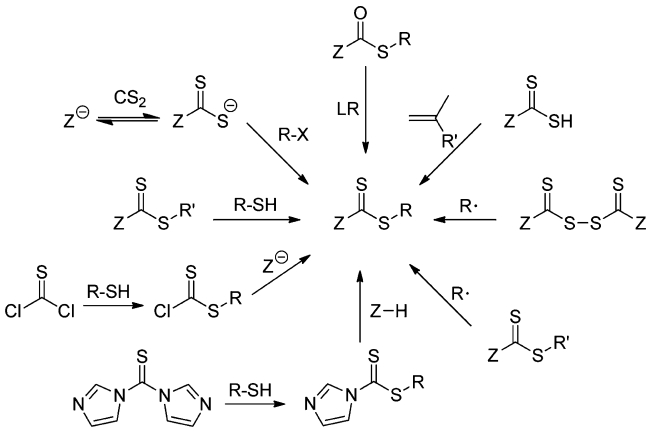
■ CONCLUSIONS AND OUTLOOK

This Perspective has reviewed the design and synthesis of RAFT agents. Many factors need to be considered when choosing a RAFT agent. The RAFT agent should be selected to achieve the anticipated molar mass, block purity, and low dispersity which requires that the reagent possess a high C_{tr}. At the same time, the RAFT agent should not retard polymerization or undergo any side reactions. It must be borne in mind that the reagents that have the highest C_{tr} are often those that are most prone to side reactions. The RAFT agent must also have appropriate solubility in the reaction medium and possess the requisite end-group functionality for the intended application. The dependence of these parameters on RAFT agent structure and the Z and R substituents is now largely understood such that a rational choice can be made.

Some guidance is available in the form of Figure 3, for selection of the Z, and Figure 4, for the selection of R.

The aforementioned factors always need to be balanced against the availability, potential toxicity, and ease of synthesis of the RAFT agent. Many methods for RAFT agent synthesis have been described. That most commonly used is the reaction of a carbodithioate salt with an alkylating agent. Other processes with mostly niche applications include various thioacylation reactions, the thiation of carbonyl compounds, the ketoform reaction, thiol exchange, and radical substitution of bis(thioacyl) disulfides and other thiocarbonylthio compounds. An attempt has been made to summarize these methods in Scheme 38.

Scheme 38. Main Methods Used for RAFT Agent Synthesis (RX = Alkylating Agent, LR = Lawesson's Reagent or Equivalent)



Challenges, however, remain in RAFT agent design and synthesis. One which has attracted the attention of several groups is the development of a truly universal RAFT agent—one able to control polymerization of all monomers polymerizable by radical polymerization. The proposal that the fluorine might be the universal Z group remains largely untested.[42,43] However, a positive outcome seems unlikely based on the experimental findings to date. Switchable RAFT agents do provide scope for controlling a much wider range of monomers.[44,45,47,107] However, there are obvious issues with the current proton switch in controlling the polymerization of monomers which possess acidic or basic functionality or where there is particular sensitivity to acid. Current work planned or underway in our laboratories seeks to address these issues by examining alternative switching mechanisms based on photochemical, redox, or other processes.

While we are some way toward a universal Z group, there is also a need for a more widely applicable or switchable R group. One possible solution lies with the single unit monomer insertion process (method M). Also needed, perhaps as a stopgap, is a convenient, high yielding, synthesis of RAFT agents with tertiary R (or equivalent) suitable for controlling the polymerization of methacrylic and other 1,1-disubstituted monomers.

However, most current activity in RAFT agent design and synthesis is in the area of functional-, macro-, and surface attached-RAFT agents with particular applications in such fields as biomedicine[158] and optoelectronics.[164] Common strategies for the preparation of macro- and surface attached-RAFT agents involve esterification or amidation of a carboxy functional RAFT agent (method I) and reactions such as azide−alkyne 1,3-dipolar cycloaddition (method J) and the active ester−amine reaction (method K). There is also continuing demand for convenient, broadly applicable, methods for end-group conversion post-RAFT polymerization,[67] and the design of the initial RAFT agent can be extremely important in facilitating these transformations—both in terms of the efficiency of the process and dealing with any byproducts that might be formed. The recent discussion on polymer chemistry and "click" reactions is extremely relevant in this context.[226]

■ AUTHOR INFORMATION

Notes

The authors declare no competing financial interest.

Biographies

Dr. Daniel Keddie obtained his PhD in synthetic organic chemistry in 2008 from the School of Physical and Chemical Sciences, Queensland University of Technology (QUT), where he worked on the synthesis of profluorescent nitroxides. He then undertook a postdoctoral position within the Institute of Health and Biomedical Innovation, QUT, involving the synthesis and characterization of amphiphilic siloxane copolymers for potential biological application. Since 2009, Dr. Keddie has held a OCE postdoctoral fellowship at the Commonwealth Scientific and Industrial Research Organization, Australia (CSIRO). He is currently working on the development of "switchable" chain transfer agents for RAFT polymerization. His research interests include free radical chemistry and the synthesis and application of functional small molecules and polymers.

Graeme Moad was born in Orange, NSW, Australia. He obtained his BSc (Hons, First Class) and PhD from the Adelaide University in the field of organic free radical chemistry. After undertaking postdoctoral research at Pennsylvania State University in the field of biological organic chemistry, he joined CSIRO in 1979 where he is is currently a chief research scientist. Dr. Moad is author or coauthor of over 150 journal papers, coinventor of 33 patent families (9 relate to the RAFT process), and coauthor of the book *The Chemistry of Radical Polymerization*. More than 12 000 papers

cite his work, and his h-index is 52. His research interests lie in the fields of polymer design and synthesis (radical polymerization, reactive extrusion, polymer nanocomposites) and polymerization kinetics and mechanism. Dr. Moad is a fellow of the Royal Australian Chemical Institute and the Australian Academy of Science. He was recently (February 2012) awarded the RACI Polymer Division's Battaerd-Jordan Medal.

Ezio Rizzardo was born in Onigo, Italy, in 1943. He graduated with First Class Honours in Applied Organic Chemistry from the University of NSW and was awarded a PhD in Organic Chemistry by the University of Sydney in 1969. He joined David Solomon's group at CSIRO in 1976 after postdoctoral work on the synthesis of biologically active compounds with Richard Turner at Rice University (Houston), Sir Derek Barton at RIMAC (Boston), and Arthur Birch at the ANU (Canberra). His research at CSIRO has focused on the development of methods for understanding and controlling polymerization processes. Research highlights from the teams he has led include radical trapping with nitroxides, nitroxide mediated polymerization, chain transfer and ring-opening polymerization by addition−fragmentation, and the RAFT process. He is coauthor of some 200 journal papers, which to date have received over 13 500 citations, and coinventor on 44 worldwide patents. He is a Fellow of the Royal Society of London, the Australian Academy of Science, the Australian Academy of Technological Sciences and Engineering and the recipient of the Australian Polymer Medal, the CSIRO Chairman's Gold Medal, and the Australian Government's Centenary Medal for contributions to society and polymer science. Dr. Rizzardo was corecipient of the Prime Minister's Prize for Science for 2011.

San Thang came to Australia as a refugee from Vietnam in 1979. He completed his BSc (Hons) in 1983 with Prof. Gus Guthrie at Griffith University and PhD also at Griffith University in 1987 with Prof. Ian Jenkins, Associate Prof. Ken Busfield, Dr. Ezio Rizzardo, and Dr. David Solomon as supervisors. He joined CSIRO in 1986 as a Research Fellow, and in late 1987, he moved to ICI Australia to undertake the challenge of industrial research in synthetic UV sunscreens and agrochemicals. San rejoined CSIRO in late 1990 and currently is a Senior Principal Research Scientist at CSIRO Materials Science and Engineering where his research focuses on the interface between biology and polymer chemistry. San has published over 100 papers in refereed journals which to date have received over 10 000 citations. He is responsible for several key inventions in the area of controlled/living radical polymerization; significantly, he is a coinventor of the RAFT process. San is a Fellow of the Australian Academy of Technological Science and Engineering and Fellow of the Royal Australian Chemical Institute. Currently, he also serves as Adjunct Professor of Monash University.

■ REFERENCES

(1) Jenkins, A. D.; Jones, R. I.; Moad, G. *Pure Appl. Chem.* **2010**, *82*, 483−491.
(2) Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2005**, *58*, 379−410.
(3) Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2006**, *59*, 669−692.
(4) Moad, G.; Rizzardo, E.; Thang, S. H. *Acc. Chem. Res.* **2008**, *41*, 1133−1142.
(5) Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2009**, *62*, 1402−1472.
(6) Moad, G.; Chiefari, J.; Krstina, J.; Postma, A.; Mayadunne, R. T. A.; Rizzardo, E.; Thang, S. H. *Polym. Int.* **2000**, *49*, 993−1001.
(7) Rizzardo, E.; Chiefari, J.; Mayadunne, R. T. A.; Moad, G.; Thang, S. H. *ACS Symp. Ser.* **2000**, *768*, 278−96.
(8) Moad, G.; Rizzardo, E.; Thang, S. H. *Polymer* **2008**, *49*, 1079−1131.
(9) Barner-Kowollik, C. *Handbook of RAFT Polymerization*; Wiley-VCH: Weinheim, Germany, 2008.
(10) Giese, B. *Radicals in Organic Synthesis: Formation of Carbon-Carbon Bonds*; Pergamon Press: Oxford, 1986.
(11) Motherwell, W. B.; Crich, D. *Free Radical Chain Reactions in Organic Synthesis*; Academic Press: London, 1992.
(12) Lewis, S. N.; Miller, J. J.; Winstein, S. *J. Org. Chem.* **1972**, *37*, 1478−1485.
(13) Keck, G. E.; Enholm, E. J.; Yates, J. B.; Wiley, M. R. *Tetrahedron* **1985**, *41*, 4079−4094.
(14) Barton, D. H. R.; McCombie, S. W. *J. Chem. Soc., Perkin Trans. 1* **1975**, 1574−1585.
(15) Cacioli, P.; Hawthorne, D. G.; Laslett, R. L.; Rizzardo, E.; Solomon, D. H. *J. Macromol. Sci., Chem.* **1986**, *A23*, 839−52.
(16) Meijs, G. F.; Rizzardo, E. *Makromol. Chem., Rapid Commun.* **1988**, *9*, 547−51.
(17) Meijs, G. F.; Rizzardo, E.; Thang, S. H. *Macromolecules* **1988**, *21*, 3122−4.
(18) Colombani, D.; Chaumont, P. *Prog. Polym. Sci.* **1996**, *21*, 439−503.
(19) Krstina, J.; Moad, C. L.; Moad, G.; Rizzardo, E.; Berge, C. T.; Fryd, M. *Macromol. Symp.* **1996**, *111*, 13−23.
(20) Moad, G.; Ercole, F.; Johnson, C. H.; Krstina, J.; Moad, C. L.; Rizzardo, E.; Spurling, T. H.; Thang, S. H.; Anderson, A. G. *ACS Symp. Ser.* **1998**, *685*, 332−60.
(21) Krstina, J.; Moad, G.; Rizzardo, E.; Winzor, C. L.; Berge, C. T.; Fryd, M. *Macromolecules* **1995**, *28*, 5381−5.
(22) Le, T. P.; Moad, G.; Rizzardo, E.; Thang, S. H. Polymerization with living characteristics. WO9801478, 1998.
(23) Chiefari, J.; Chong, Y. K.; Ercole, F.; Krstina, J.; Jeffery, J.; Le, T. P. T.; Mayadunne, R. T. A.; Meijs, G. F.; Moad, C. L.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromolecules* **1998**, *31*, 5559−62.

(24) Corpart, P.; Charmot, D.; Biadatti, T.; Zard, S.; Michelet, D. Block polymer synthesis by controlled radical polymerization. WO9858974, 1998.
(25) Charmot, D.; Corpart, P.; Adam, H.; Zard, S. Z.; Biadatti, T.; Bouhadir, G. *Macromol. Symp.* **2000**, *150*, 23−32.
(26) Barner-Kowollik, C.; Buback, M.; Charleux, B.; Coote, M. L.; Drache, M.; Fukuda, T.; Goto, A.; Klumperman, B.; Lowe, A. B.; Mcleary, J. B.; Moad, G.; Monteiro, M. J.; Sanderson, R. D.; Tonge, M. P.; Vana, P. *J. Polym. Sci., Part A: Polym. Chem.* **2006**, *44*, 5809−5831.
(27) Klumperman, B.; van den Dungen, E. T. A.; Heuts, J. P. A.; Monteiro, M. J. *Macromol. Rapid Commun.* **2010**, *31*, 1846−1862.
(28) Meiser, W.; Barth, J.; Buback, M.; Kattner, H.; Vana, P. *Macromolecules* **2011**, *44*, 2474−2480.
(29) Junkers, T. *J. Polym. Sci., Part A: Polym. Chem.* **2011**, *49*, 4154−4163.
(30) Ting, S. R. S.; Davis, T. P.; Zetterlund, P. B. *Macromolecules* **2011**, *44*, 4187−4193.
(31) Moad, G.; Barner-Kowollik, C. The Mechanism and Kinetics of the RAFT Process: Overview, Rates, Stabilities, Side Reactions, Product Spectrum and Outstanding Challenges. In *Handbook of RAFT Polymerization*; Barner-Kowollik, C., Ed.; Wiley-VCH: Weinheim, Germany, 2008; pp 51−104.
(32) Moad, G.; Chiefari, J.; Moad, C. L.; Postma, A.; Mayadunne, R. T. A.; Rizzardo, E.; Thang, S. H. *Macromol. Symp.* **2002**, *182*, 65−80.
(33) Chong, Y. K.; Krstina, J.; Le, T. P. T.; Moad, G.; Postma, A.; Rizzardo, E.; Thang, S. H. *Macromolecules* **2003**, *36*, 2256−2272.
(34) Chiefari, J.; Mayadunne, R. T. A.; Moad, C. L.; Moad, G.; Rizzardo, E.; Postma, A.; Skidmore, M. A.; Thang, S. H. *Macromolecules* **2003**, *36*, 2273−2283.
(35) Müller, A. H. E.; Litvenko, G. *Macromolecules* **1997**, *30*, 1253−66.
(36) Müller, A. H. E.; Zhuang, R.; Yan, D.; Litvenko, G. *Macromolecules* **1995**, *28*, 4326−33.
(37) Coote, M. L.; Henry, D. J. *Macromolecules* **2005**, *38*, 1415−1433.
(38) Coote, M. L.; Krenske, E. H.; Izgorodina, E. I. Quantum-chemical studies of RAFT polymerization: methodology, structure-reactivity correlations and kinetic implications. In *Handbook of RAFT Polymerization*; Barner-Kowollik, C., Ed.; Wiley-VCH: Weinheim, Germany, 2008; pp 5−49.
(39) Destarac, M.; Bzducha, W.; Taton, D.; Gauthier-Gillaizeau, I.; Zard, S. Z. *Macromol. Rapid Commun.* **2002**, *23*, 1049−1054.
(40) Mayadunne, R. T. A.; Rizzardo, E.; Chiefari, J.; Chong, Y. K.; Moad, G.; Thang, S. H. *Macromolecules* **1999**, *32*, 6977−6980.
(41) Destarac, M.; Charmot, D.; Franck, X.; Zard, S. Z. *Macromol. Rapid Commun.* **2000**, *21*, 1035−1039.
(42) Coote, M. L.; Izgorodina, E. I.; Cavigliasso, G. E.; Roth, M.; Busch, M.; Barner-Kowollik, C. *Macromolecules* **2006**, *39*, 4585−4591.
(43) Theis, A.; Stenzel, M. H.; Davis, T. P.; Coote, M. L.; Barner-Kowollik, C. *Aust. J. Chem.* **2005**, *58*, 437−441.
(44) Benaglia, M.; Chiefari, J.; Chong, Y. K.; Moad, G.; Rizzardo, E.; Thang, S. H. *J. Am. Chem. Soc.* **2009**, *131*, 6914−6915.
(45) Benaglia, M.; Chen, M.; Chong, Y. K.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromolecules* **2009**, *42*, 9384−9386.
(46) Moad, G.; Benaglia, M.; Chen, M.; Chiefari, J.; Chong, Y., K.; Keddie, D., J.; Rizzardo, E.; Thang San, H. *ACS Symp. Ser.* **2011**, *1066*, 81−102.
(47) Keddie, D. J.; Guerrero-Sanchez, C.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromolecules* **2011**, *44*, 6738−6745.
(48) Keddie, D. J.; Guerrero-Sanchez, C.; Moad, G.; Mulder, R.; Rizzardo, E.; Thang, S. H. *Macromolecules* **2012**; DOI: 10.1021/ma300616g.
(49) Stenzel, M. H.; Cummins, L.; Roberts, G. E.; Davis, T. R.; Vana, P.; Barner-Kowollik, C. *Macromol. Chem. Phys.* **2003**, *204*, 1160−1168.
(50) Coote, M. L.; Radom, L. *Macromolecules* **2004**, *37*, 590−596.
(51) Bicciocchi, E.; Chong, Y. K.; Giorgini, L.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromol. Chem. Phys.* **2010**, *211*, 529−538.
(52) Nebhani, L.; Sinnwell, S.; Lin, C. Y.; Coote, M. L.; Stenzel, M. H.; Barner-Kowollik, C. *J. Polym. Sci., Part A: Polym. Chem.* **2009**, *47*, 6053−6071.
(53) Thomas, D. B.; Convertine, A. J.; Hester, R. D.; Lowe, A. B.; McCormick, C. L. *Macromolecules* **2004**, *37*, 1735−1741.
(54) Metzner, P. *Top. Curr. Chem.* **1999**, *204*, 127−181.
(55) Chen, M.; Moad, G.; Rizzardo, E. *Aust. J. Chem.* **2011**, *64*, 433−437.
(56) Moad, C. L.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromolecules* **1996**, *29*, 7717−26.
(57) Rizzardo, E.; Chen, M.; Chong, B.; Moad, G.; Skidmore, M.; Thang, S. H. *Macromol. Symp.* **2007**, *248*, 104−116.
(58) Li, C. Z.; Benicewicz, B. C. *J. Polym. Sci., Part A: Polym. Chem.* **2005**, *43*, 1535−1543.
(59) Fischer, H.; Radom, L. *Angew. Chem., Int. Ed.* **2001**, *40*, 1340−71.
(60) Moad, G.; Rizzardo, E.; Thang, S. H. *Mater. Matters (Milwaukee, WI, U. S.)* **2010**, 2−5.
(61) Moad, G.; Rizzardo, E.; Thang, S. H. *Strem Chem.* **2011**, *25*, 2−10.
(62) Destarac, M. *Macromol. React. Eng.* **2010**, *4*, 165−179.
(63) Couvreur, L.; Guerret, O.; Laffitte, J.-A.; Magnet, S. *Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.)* **2005**, *46*, 219−220.
(64) Brzytwa, A. J.; Johnson, J. *Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem)* **2011**, *52* (2), 533−534.
(65) Willcock, H.; O'Reilly, R. K. *Polym. Chem.* **2010**, *1*, 149−157.
(66) Moad, G.; Chong, Y. K.; Rizzardo, E.; Postma, A.; Thang, S. H. *Polymer* **2005**, *46*, 8458−8468.
(67) Moad, G.; Rizzardo, E.; Thang, S. H. *Polym. Int.* **2011**, *60*, 9−25.
(68) Barner, L.; Perrier, S. Polymers with well-defined end groups via RAFT - synthesis, applications and postmodifications. In *Handbook of RAFT Polymerization*; Barner-Kowollik, C., Ed.; Wiley-VCH: Weinheim, Germany, 2008; pp 455−482.
(69) Chong, B.; Moad, G.; Rizzardo, E.; Skidmore, M.; Thang, S. H. *Aust. J. Chem.* **2006**, *59*, 755−762.
(70) Postma, A.; Davis, T. P.; Evans, R. A.; Li, G.; Moad, G.; O'Shea, M. *Macromolecules* **2006**, *39*, 5293−5306.
(71) Postma, A.; Davis, T. P.; Moad, G.; O'Shea, M. S. *Macromolecules* **2005**, *38*, 5371−5374.
(72) Postma, A.; Davis, T. P.; Moad, G.; O'Shea, M. S. *React. Funct. Polym.* **2006**, *66*, 137−147.
(73) Chong, Y. K.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromolecules* **2007**, *40*, 4446−4455.
(74) Perrier, S.; Takolpuckdee, P.; Mars, C. A. *Macromolecules* **2005**, *38*, 2033−2036.
(75) Chen, M.; Moad, G.; Rizzardo, E. *J. Polym. Sci., Part A: Polym. Chem* **2009**, *47*, 6704−6714.
(76) Moad, G.; Thang, S. H.; Rizzardo, E. Process for transforming the end groups of polymers. WO08103144, 20080828, 2008.
(77) Purvis, J. E.; Jones, H. O.; Tasker, H. S. *J. Chem. Soc., Trans.* **1910**, *97*, 2287−97.
(78) Bost, R. W.; Mattox, W. J. *J. Am. Chem. Soc.* **1930**, *52*, 332−335.
(79) Bost, R. W.; Williams, W. W. *J. Am. Chem. Soc.* **1930**, *52*, 4991−4992.
(80) Mayer, R.; Scheithauer, S. Dithiocarbonsäuren, deren salze and ester. In *Houben - Weyl Methods of Organic Chemistry*; Hanack, M., Ed.; Thieme: Stuttgart, 1985; Vol. *E*, pp 891−930.
(81) Kato, S.; Ishida, M. *Sulfur Rep.* **1988**, *8*, 155−323.
(82) Levesque, G.; Arsène, P.; Fanneau-Bellenger, V.; Pham, T.-N. *Biomacromolecules* **2000**, *1*, 387−399.
(83) Glass, R. S. *Sci. Synth* **2005**, *22*, 109−132.
(84) Rizzardo, E.; Moad, G.; Thang, S. H. RAFT Polymerization in Bulk Monomer or in (Organic) Solution. In *Handbook of RAFT Polymerization*; Barner-Kowollik, C., Ed.; Wiley-VCH: Weinheim, Germany, 2008; pp 189−234.
(85) Bonnans-Plaisance, C.; Gressier, J. C.; Levesque, G.; Mahjoub, A. *Bull. Soc. Chim. Fr.* **1985**, 891−9.
(86) Ladaviere, C.; Dorr, N.; Claverie, J. P. *Macromolecules* **2001**, *34*, 5370−5372.

(87) Skey, J.; O'Reilly, R. K. *Chem. Commun.* **2008**, 4183−4185.
(88) Haridharan, N.; Dhamodharan, R. *J. Polym. Sci., Part A: Polym. Chem* **2011**, *49*, 1021−1032.
(89) Haridharan, N.; Ponnusamy, K.; Dhamodharan, R. *J. Polym. Sci., Part A: Polym. Chem.* **2010**, *48*, 5329−5338.
(90) Schmidt, B. V. K. J.; Hetzer, M.; Ritter, H.; Barner-Kowollik, C. *Macromolecules* **2011**, *44*, 7220−7232.
(91) Meijer, J.; Vermeer, P.; Brandsma, L. *Recueil* **1973**, *92*, 601−604.
(92) Di Carlo, G.; Damin, F.; Armelao, L.; Maccato, C.; Unlu, S.; Spuhler, P. S.; Chiari, M. *Appl. Surf. Sci.* **2012**, *258*, 3750−3756.
(93) Kessler, D.; Theato, P. *Macromolecules* **2008**, *41*, 5237−5244.
(94) Legge, T. M.; Slark, A. T.; Perrier, S. *Macromolecules* **2007**, *40*, 2318−2326.
(95) Becke, F.; Hagen, H. Dithiobenzoic acids. US 3636089, 1972.
(96) Le, T. P.; Moad, G.; Rizzardo, E.; Thang, S. H. Polymerization with living characteristics. US7714075, 2010.
(97) Cohen, I. A.; Basolo, F. *Inorg. Chem.* **1964**, *3*, 1641−1642.
(98) March, J. *Advanced Organic Chemistry*, 3rd ed.; John Wiley & Sons: New York, 1985.
(99) Hu, D.; Zheng, S. *Polymer* **2010**, *51*, 6346−6354.
(100) Xue, X.; Zhu, J.; Zhang, Z.; Cheng, Z.; Tu, Y.; Zhu, X. *Polymer* **2010**, *51*, 3083−3090.
(101) Chen, K.; Grant, N.; Liang, L.; Zhang, H.; Tan, B. *Macromolecules* **2010**, *43*, 9355−9364.
(102) Stamenović, M. M.; Espeel, P.; Camp, W. V.; Du Prez, F. E. *Macromolecules* **2011**, *44*, 5619−5630.
(103) Nagai, A.; Yoshii, R.; Otsuka, T.; Kokado, K.; Chujo, Y. *Langmuir* **2010**, *26*, 15644−15649.
(104) Jana, S.; Parthiban, A. *Macromol. Chem. Phys.* **2011**, *212*, 790−798.
(105) Quemener, D.; Davis, T. P.; Barner-Kowollik, C.; Stenzel, M. H. *Chem. Commun.* **2006**, 5051−5053.
(106) O'Reilly, R. Personal Communication, 2011. The base for this reaction was caesium carbonate not potassium phosphate as stated in the original publication (Electronic Supplementary Information to ref 87).
(107) Moad, G.; Benaglia, M.; Chen, M.; Chiefari, J.; Chong, Y., K.; Keddie, D., J.; Rizzardo, E.; Thang San, H. *ACS Symp. Ser.* **2011**, *1066*, 81−102.
(108) Lee, A. W. M.; Chan, W. H.; Wong, H. C. *Synth. Commun.* **1988**, *18*, 1531−1536.
(109) Weeldenburg, J. G. *Recl. Trav. Chim. Pays-Bas* **1928**, *47*, 496−512.
(110) Wertheim, E. *J. Am. Chem. Soc.* **1926**, *48*, 826−830.
(111) Aoyagi, N.; Ochiai, B.; Mori, H.; Endo, T. *Synlett* **2006**, 636−638.
(112) Aoyagi, N.; Endo, T. *J. Polym. Sci., Part A: Polym. Chem.* **2009**, *47*, 3702−3709.
(113) Degani, I.; Fochi, R.; Gatti, A.; Regondi, V. *Synthesis* **1986**, 894−9.
(114) Majonis, D.; Herrera, I.; Ornatsky, O.; Schulze, M.; Lou, X.; Soleimani, M.; Nitz, M.; Winnik, M. A. *Anal. Chem.* **2010**, *82*, 8961−8969.
(115) Soleiman-Beigi, M.; Arzehgar, Z.; Movassagh, B. *Synthesis* **2010**, 392−394.
(116) Kiasat, A. R.; Badri, R.; Sayyahi, S. *Chin. Chem. Lett.* **2008**, *19*, 1301−1304.
(117) Haynes, W. M. *CRC Handbook of Chemistry and Physics*, 92nd ed.; CRC Press: Boca Raton, FL, 2012.
(118) Chiefari, J.; Mayadunne, R. T.; Moad, G.; Rizzardo, E.; Thang, S. H. Polymerization process with living characteristics and polymers made therefrom. US6747111, 2004.
(119) Wood, M. R.; Duncalf, D. J.; Rannard, S. P.; Perrier, S. *Org. Lett.* **2006**, *8*, 553−556.
(120) Wazneh, L.; Guillemin, J. C.; Guenot, P.; Vallée, Y.; Denis, J. M. *Tetrahedron Lett.* **1988**, *29*, 5899−5900.
(121) Wepplo, P. *Synth. Commun.* **1989**, *19*, 1533−8.
(122) Mathias, E.; Shimanski, M. *J. Chem. Soc., Chem. Commun.* **1981**, 569−70.
(123) Lecher, H. Z.; Greenwood, R. A.; Whitehouse, K. C.; Chao, T. H. *J. Am. Chem. Soc.* **1956**, *78*, 5018−5022.
(124) Davy, H. *J. Chem. Soc., Chem. Commun.* **1982**, 457−458.
(125) Jesberger, M.; Barner, L.; Stenzel, M. H.; Malmstrom, E.; Davis, T. P.; Barner-Kowollik, C. *J. Polym. Sci., Part A: Polym. Chem.* **2003**, *41*, 3847−3861.
(126) Kanagasabapathy, S.; Sudalai, A.; Benicewicz, B. C. *Tetrahedron Lett.* **2001**, *42*, 3791−3794.
(127) Sudalai, A.; Kanagasabapathy, S.; Benicewicz, B. C. *Org. Lett.* **2000**, *2*, 3213−3216.
(128) Dureault, A.; Taton, D.; Destarac, M.; Leising, F.; Gnanou, Y. *Macromolecules* **2004**, *37*, 5513−5519.
(129) Dureault, A.; Gnanou, Y.; Taton, D.; Destarac, M.; Leising, F. *Angew. Chem., Int. Ed.* **2003**, *42*, 2869−2872.
(130) Lai, J. T. *Tetrahedron Lett.* **2002**, *43*, 1965−1967.
(131) Lai, J. T.; Filla, D.; Shea, R. *Macromolecules* **2002**, *35*, 6754−6756.
(132) Lai, J. T.; Shea, R. *J. Polym. Sci., Part A: Polym. Chem.* **2006**, *44*, 4298−4316.
(133) Harrisson, S.; Wooley, K. L. *Chem. Commun.* **2005**, 3259−3261.
(134) Chikhaoui-Grioune, D.; Aqil, A.; Zalfen, A. M.; Benaboura, A.; Jerome, C. *J. Appl. Polym. Sci.* **2010**, *117*, 1005−1012.
(135) You, Y.-Z.; Kalebaila, K. K.; Brock, S. L.; Oupicky, D. *Chem. Mater.* **2008**, *20*, 3354−3359.
(136) Zhang, T.; Wu, Y.; Pan, X.; Zheng, Z.; Ding, X.; Peng, Y. *Eur. Polym. J.* **2009**, *45*, 1625−1633.
(137) Liu, Z.; Hu, J.; Sun, J.; He, G.; Li, Y.; Zhang, G. *J. Polym. Sci., Part A: Polym. Chem.* **2010**, *48*, 3573−3586.
(138) Yan, L.; Tao, W. *Polymer* **2010**, *51*, 2161−2167.
(139) Yan, Q.; Yuan, J.; Kang, Y.; Cai, Z.; Zhou, L.; Yin, Y. *Chem. Commun. (Cambridge, U. K.)* **2010**, *46*, 2781−2783.
(140) Yodice, R.; Johnson, J. R.; Beebe, R. L.; Brzytwa, A. J.; Hilker, C. D. Preparation of substituted trithiocarbonate derivatives by one-step process. WO2009035793A1, 2009.
(141) Leon, N. H.; Asquith, R. S. *Tetrahedron* **1970**, *26*, 1719−1725.
(142) Severac, R.; Lacroix-Desmazes, P.; Boutevin, B. *Polym. Int.* **2002**, *51*, 1117−1122.
(143) Favier, A.; Charreyre, M. T.; Chaumont, P.; Pichot, C. *Macromolecules* **2002**, *35*, 8271−8280.
(144) Chernikova, E.; Terpugova, P.; Bui, C. O.; Charleux, B. *Polymer* **2003**, *44*, 4101−4107.
(145) Oae, S.; Yagihara, T.; Okabe, T. *Tetrahedron* **1972**, *28*, 3203−16.
(146) Barner-Kowollik, C.; Quinn, J. F.; Nguyen, T. L. U.; Heuts, J. P. A.; Davis, T. P. *Macromolecules* **2001**, *34*, 7849−7857.
(147) Merican, Z.; Schiller, T. L.; Hawker, C. J.; Fredericks, P. M.; Blakey, I. *Langmuir* **2007**, *23*, 10539−10545.
(148) Plummer, R.; Goh, Y. K.; Whittaker, A. K.; Monteiro, M. J. *Macromolecules* **2005**, *38*, 5352−5355.
(149) Thang, S. H.; Chong, Y. K.; Mayadunne, R. T. A.; Moad, G.; Rizzardo, E. *Tetrahedron Lett.* **1999**, *40*, 2435−8.
(150) Bouhadir, G.; Legrand, N.; Quiclet-Sire, B.; Zard, S. Z. *Tetrahedron Lett.* **1999**, *40*, 277−80.
(151) Weber, W. G.; McLeary, J. B.; Sanderson, R. D. *Tetrahedron Lett.* **2006**, *47*, 4771−4774.
(152) Mitsukami, Y.; Donovan, M. S.; Lowe, A. B.; McCormick, C. L. *Macromolecules* **2001**, *34*, 2248−2256.
(153) Roth, P. J.; Wiss, K. T.; Zentel, R.; Theato, P. *Macromolecules* **2008**, *41*, 8513−8519.
(154) Wager, C. M.; Haddleton, D. M.; Bon, S. A. F. *Eur. Polym. J.* **2004**, *40*, 641−645.
(155) Kwak, Y.; Nicolay, R.; Matyjaszewski, K. *Macromolecules* **2009**, *42*, 3738−3742.
(156) Alberti, A.; Benaglia, M.; Laus, M.; Sparnacci, K. *J. Org. Chem.* **2002**, *67*, 7911−7914.
(157) Benaglia, M.; Rizzardo, E.; Alberti, A.; Guerra, M. *Macromolecules* **2005**, *38*, 3129−3140.

(158) Boyer, C.; Bulmus, V.; Davis, T. P.; Ladmiral, V.; Liu, J.; Perrier, S. *Chem. Rev.* **2009**, *109*, 5402−5436.
(159) Broyer, R. M.; Grover, G. N.; Maynard, H. D. *Chem. Commun.* **2011**, *47*, 2212−2226.
(160) Dehn, S.; Chapman, R.; Jolliffe, K. A.; Perrier, S. *Polym. Rev.* **2011**, *51*, 214−234.
(161) Barz, M.; Luxenhofer, R.; Zentel, R.; Vicent, M. J. *Polym. Chem.* **2011**, *2*, 1900−1918.
(162) Chu, D. S. H.; Schellinger, J. G.; Shi, J.; Convertine, A. J.; Stayton, P. S.; Pun, S. H. *Acc. Chem. Res.* **2012**; DOI: 10.1021/ar200242z.
(163) Sumerlin, B. S. *ACS Macro Lett.* **2011**, *1*, 141−145.
(164) Moad, G.; Chen, M.; Häussler, M.; Postma, A.; Rizzardo, E.; Thang, S. H. *Polym. Chem.* **2011**, *2*, 492−519.
(165) Chen, M.; Haeussler, M.; Moad, G.; Rizzardo, E. *Org. Biomol. Chem.* **2011**, *9*, 6111−6119.
(166) Rzayev, J.; Hillmyer, M. A. *Macromolecules* **2004**, *38*, 3−5.
(167) Rzayev, J.; Hillmyer, M. A. *J. Am. Chem. Soc.* **2005**, *127*, 13373−13379.
(168) Pai, T. S. C.; Barner-Kowollik, C.; Davis, T. P.; Stenzel, M. H. *Polymer* **2004**, *45*, 4383−4389.
(169) Shi, L.; Chapman, T. M.; Beckman, E. J. *Macromolecules* **2003**, *36*, 2563−2567.
(170) Hentschel, J.; Bleek, K.; Ernst, O.; Lutz, J. F.; Borner, H. G. *Macromolecules* **2008**, *41*, 1073−1075.
(171) ten Cate, M. G. J.; Rettig, H.; Bernhardt, K.; Borner, H. G. *Macromolecules* **2005**, *38*, 10643−10649.
(172) Zheng, Q.; Zheng, G.-H.; Pan, C.-Y. *Polym. Int.* **2006**, *55*, 1114−1123.
(173) Yuan, K.; Lu, L.-L.; Li, Z.-F.; Shi, X.-N. *Chin. J. Chem.* **2008**, *26*, 1929−1934.
(174) Briquel, R.; Mazzolini, J.; Bris, T. L.; Boyron, O.; Boisson, F.; Delolme, F.; D'Agosto, F.; Boisson, C.; Spitz, R. *Angew. Chem., Int. Ed.* **2008**, *47*, 9311−9313.
(175) Aamer, K. A.; Tew, G. N. *J. Polym. Sci., Part A: Polym. Chem.* **2007**, *45*, 5618−5625.
(176) Li, C.; Han, J.; Ryu, C. Y.; Benicewicz, B. C. *Macromolecules* **2006**, *39*, 3175−3183.
(177) Gondi, S. R.; Vogt, A. P.; Sumerlin, B. S. *Macromolecules* **2007**, *40*, 474−481.
(178) Liu, J.; Liu, H.; Boyer, C.; Bulmus, V.; Thomas, P.; Davis. *J. Polym. Sci., Part A: Polym. Chem.* **2009**, *47*, 899−912.
(179) Ranjan, R.; Brittain, W. J. *Macromol. Rapid Commun.* **2007**, *28*, 2084−2089.
(180) Vora, A.; Singh, K.; Webster, D. C. *Polymer* **2009**, *50*, 2768−2774.
(181) Zhang, X.; Li, J.; Li, W.; Zhang, A. *Biomacromolecules* **2007**, *8*, 3557−3567.
(182) De, P.; Li, M.; Gondi, S. R.; Sumerlin, B. S. *J. Am. Chem. Soc.* **2008**, *130*, 11288−11289.
(183) Vestberg, R.; Piekarski, A. M.; Pressly, E. D.; Berkel, K. Y. V.; Malkoch, M.; Gerbac, J.; Ueno, N.; Hawker, C. J. *J. Polym. Sci., Part A: Polym. Chem.* **2009**, *47*, 1237−1258.
(184) Chong, Y. K.; Le, T. P. T.; Moad, G.; Rizzardo, E.; Thang, S. H. *Macromolecules* **1999**, *32*, 2071−4.
(185) Chen, M.; Moad, G.; Rizzardo, E.; Evans, R. A.; Haeussler, M. Conducting and Semiconducting Organic Materials. WO2009155657A1, 2009.
(186) Bathfield, M.; D'Agosto, F.; Spitz, R.; Charreyre, M. T.; Delair, T. *J. Am. Chem. Soc.* **2006**, *128*, 2546−2547.
(187) Liu, J.; Yang, W.; Zareie, H. M.; Gooding, J. J.; Davis, T. P. *Macromolecules* **2009**, *42*, 2931−2939.
(188) Evans, R. A. *Aust. J. Chem.* **2007**, *60*, 384−395.
(189) Iha, R. K.; Wooley, K. L.; Nystrom, A. M.; Burke, D. J.; Kade, M. J.; Hawker, C. J. *Chem. Rev.* **2009**, *109*, 5620−5686.
(190) Binder, W. H.; Sachsenhofer, R. *Macromol. Rapid Commun.* **2007**, *28*, 15−54.
(191) Binder, W. H.; Sachsenhofer, R. *Macromol. Rapid Commun.* **2008**, *29*, 952−981.
(192) Sumerlin, B. S.; Vogt, A. P. *Macromolecules* **2009**, *43*, 1−13.
(193) Golas, P. L.; Matyjaszewski, K. *Chem. Soc. Rev.* **2010**, *39*, 1338−1354.
(194) Ladmiral, V.; Legge, T. M.; Zhao, Y. L.; Perrier, S. *Macromolecules* **2008**, *41*, 6728−6732.
(195) Akeroyd, N.; Pfukwa, R.; Klumperman, B. *Macromolecules* **2009**, *42*, 3014−3018.
(196) Hoyle, C. E.; Lowe, A. B.; Bowman, C. N. *Chem. Soc. Rev.* **2010**, *39*, 1355−1387.
(197) Li, M.; Li, H.; De, P.; Sumerlin, B. S. *Macromol. Rapid Commun.* **2011**, *32*, 354−359.
(198) Boyer, C.; Bulmus, V.; Liu, J. Q.; Davis, T. P.; Stenzel, M. H.; Barner-Kowollik, C. *J. Am. Chem. Soc.* **2007**, *129*, 7145−7154.
(199) Pound, G.; McLeary, J. B.; McKenzie, J. M.; Lange, R. F. M.; Klumperman, B. *Macromolecules* **2006**, *39*, 7796−7797.
(200) McLeary, J. B.; Tonge, M. P.; Klumperman, B. *Macromol. Rapid Commun.* **2006**, *27*, 1233−1240.
(201) McLeary, J. B.; McKenzie, J. M.; Tonge, M. P.; Sanderson, R. D.; Klumperman, B. *Chem. Commun.* **2004**, 1950−1951.
(202) McLeary, J. B.; Calitz, F. M.; McKenzie, J. M.; Tonge, M. P.; Sanderson, R. D.; Klumperman, B. *Macromolecules* **2005**, *38*, 3151−3161.
(203) McLeary, J. B.; Calitz, F. M.; McKenzie, J. M.; Tonge, M. P.; Sanderson, R. D.; Klumperman, B. *Macromolecules* **2004**, *37*, 2383−2394.
(204) Quiclet-Sire, B.; Zard, S. Z. *Org. Lett.* **2008**, *10*, 3279−3282.
(205) Quiclet-Sire, B.; Zard, S. Z. *Pure Appl. Chem.* **2011**, *83*, 519−551.
(206) Chen, M.; Ghiggino, K. P.; Mau, A. W. H.; Rizzardo, E.; Sasse, W. H. F.; Thang, S. H.; Wilson, G. J. *Macromolecules* **2004**, *37*, 5479−5481.
(207) Destarac, M.; Blidi, I.; Coutelier, O.; Guinaudeau, A.; Mazières, S.; van Gramberen, E.; Wilson, J. Aqueous RAFT/MADIX Polymerization: Same Monomers, New Polymers? In *Progress in Controlled Radical Polymerization: Mechanisms and Techniques*; American Chemical Society: Washington, DC, 2012; Vol. *1100*, pp 259−275.
(208) Chen, M.; Ghiggino, K. P.; Rizzardo, E.; Thang, S. H.; Wilson, G. J. *Chem. Commun.* **2008**, 1112−1114.
(209) Tanaka, S.; Nishida, H.; Endo, T. *Macromolecules* **2009**, *42*, 293−298.
(210) Houshyar, S.; Keddie, D.; Moad, G.; Mulder, R.; Saubern, S.; Tsanaktsidis, J. *Polym. Chem* **2012**, DOI: 10.1039/C2PY00529H.
(211) Chaffey-Millar, H.; Stenzel, M. H.; Davis, T. P.; Coote, M. L.; Barner-Kowollik, C. *Macromolecules* **2006**, *39*, 6406−6419.
(212) Li, Y.; Schadler, L. S.; Benicewicz, B. C. Surface and particle modification via the RAFT process: approach and properties. In *Handbook of RAFT Polymerization*; Barner-Kowollik, C., Ed.; Wiley-VCH: Weinheim, Germany, 2008; pp 423−453.
(213) Zhang, B.; Chen, Y.; Xu, L.; Zeng, L.; He, Y.; Kang, E.-T.; Zhang, J. *J. Polym. Sci., Part A: Polym. Chem.* **2011**, *49*, 2043−2050.
(214) Semsarilar, M.; Ladmiral, V.; Perrier, S. *J. Polym. Sci., Part A: Polym. Chem.* **2010**, *48*, 4361−4365.
(215) Golden, A. L.; Battrell, C. F.; Pennell, S.; Hoffman, A. S.; J. Lai, J.; Stayton, P. S. *Bioconjugate Chem.* **2010**, *21*, 1820−1826.
(216) Bhattacharjee, S.; Bong, D. *J. Polym. Environ.* **2011**, *19*, 203−208.
(217) Ranjan, R.; Brittain, W. J. *Macromolecules* **2007**, *40*, 6217−6223.
(218) Ranjan, R.; Brittain, W. J. *Macromol. Rapid Commun.* **2008**, *29*, 1104−1110.
(219) Lim, J.; Yang, H.; Paek, K.; Cho, C.-H.; Kim, S.; Bang, J.; Kim, B. J. *J. Polym. Sci., Part A: Polym. Chem.* **2011**, *49*, 3464−3474.
(220) Pfaff, A.; Barner, L.; Müller, A. H. E.; Granville, A. M. *Eur. Polym. J.* **2011**, *47*, 805−815.
(221) Grande, C. D.; Tria, M. C.; Jiang, G.; Ponnapati, R.; Park, Y.; Zuluaga, F.; Advincula, R. *React. Funct. Polym.* **2011**, *71*, 938−942.

(222) Grande, C. D.; Tria, M. C.; Jiang, G.; Ponnapati, R.; Advincula, R. *Macromolecules* **2011**, *44*, 966−975.
(223) Tria, M. C. R.; Advincula, R. C. *Macromol. Rapid Commun.* **2011**, *32*, 966−971.
(224) Roohi, F.; Titirici, M. M. *New J. Chem.* **2008**, *32*, 1409−1414.
(225) Zhou, J.; Li, W.; Gu, J.-S.; Yu, H.-Y.; Tang, Z.-Q.; Wei, X.-W. *Sep. Purif. Technol.* **2010**, *71*, 233−240.
(226) Barner-Kowollik, C.; Du Prez, F. E.; Espeel, P.; Hawker, C. J.; Junkers, T.; Schlaad, H.; Van Camp, W. *Angew. Chem., Int. Ed.* **2011**, *50*, 60−62.